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The Π_1 -Nondefinability of NS_{ω_1}

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The Π_1 -Nondefinability of NS_{ω_1}

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Abstract

NS_{ω_1} , the nonstationary ideal on ω_1 , is naturally Σ_1 -definable over H_{ω_2} using ω_1 as the only parameter. It has been known that certain set theoretic statements are compatible with a Π_1 -definition of NS_{ω_1} modulo reasonable assumptions. In the present thesis, we prove that if BMM holds true and NS_{ω_1} is saturated, then NS_{ω_1} is not boldface Π_1 -definable over H_{ω_2} . One of the main techniques of the proof of this result is a modified version of Asperó–Schindler forcing. The details of this forcing are given. We introduce the uB-projective hierarchy of subsets of the generalized Baire space ${}^{\omega_1}\omega_1$ and prove that if Γ^∞ -BMM holds true and there is a Woodin cardinal, then NS_{ω_1} is not uB- Π_1^1 . We also introduce the concept and investigate the properties of Σ_1 -completeness for subsets of the generalized Cantor space ${}^{\omega_1}2$.

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Chapter 1

Overview

Let κ be a regular uncountable cardinal. The nonstationary ideal on κ , denoted by NS_κ , is naturally Σ_1 -definable over H_{κ^+} using κ as the only parameter:

$$\text{NS}_\kappa = \{A \subset \kappa : \exists C (C \text{ is club in } \kappa \text{ and } C \cap A = \emptyset)\}.$$

We ask, however, if it is possible to define NS_κ in a Π_1 fashion over H_{κ^+} using κ or a subset of κ as a parameter; or equivalently, if there is a Σ_1 -formula φ and a set $X \subset \kappa$ such that for all $S \subset \kappa$, S is stationary if and only if $\varphi(S, X)$. It turns out this question in general cannot be answered within ZFC. The current research interest is to determine if a Π_1 -definition for NS_{ω_1} is compatible with common set theoretic principles, e.g. forcing axioms. We briefly review some known results. We start with results which negatively answer the above question.

- If $V = L$, then for all regular uncountable κ , NS_κ is *not* Π_1 -definable ([10]).¹
- If κ is a measurable cardinal, then NS_κ is *not* Π_1 -definable ([12]).
- If Bounded Martin's Maximum (abbr. BMM) holds true and there is a Woodin cardinal, then NS_{ω_1} is *not* Π_1 -definable ([15]).
- If Axiom (*) holds true, then NS_{ω_1} is *not* Π_1 -definable ([15]);

hence by Asperó–Schindler's theorem that Martin's Maximum⁺⁺ (abbr. MM^{++}) implies Axiom (*) ([3]),

¹In the present thesis, by “ Π_1 -definable” we always mean “boldface Π_1 -definable over H_{κ^+} ” where κ is the regular uncountable cardinal on which the nonstationary ideal is being talked about.

- if MM^{++} holds true, then NS_{ω_1} is *not* Π_1 -definable.

It is, however, consistent to have a Π_1 -definition for the nonstationary ideal together with common set theoretic statements modulo reasonable assumptions, e.g. large cardinals:

- If $V = L$, then for every successor cardinal κ , there is a cardinal preserving forcing in whose generic extension NS_κ is Π_1 -definable ([11]).
- If there is a measurable cardinal, then in a forcing extension of V , NS_{ω_1} is precipitous and Π_1 -definable ([12]).
- If M_1 exists, then in a forcing extension of M_1 , NS_{ω_1} is saturated and NS_{ω_1} is Π_1 -definable with parameter ω_1 ([14]).²
- If there is a Woodin cardinal, then in a forcing extension of V , NS_{ω_1} is saturated and Π_1 -definable ([14]).
- If $V = L$ and there is a reflecting cardinal, then in a forcing extension of L , BPFA holds true and NS_{ω_1} is Π_1 -definable ([15]).

The same authors of [15] also showed that BPFA as in the above item can be improved to PFA if stronger large cardinals are assumed:

- if there is a supercompact cardinal, then in a forcing extension of V , PFA holds true and NS_{ω_1} is Π_1 -definable ([16]).

Strong Reflection Principle, abbr. SRP, a combinatorial principle that is implied by MM and implies a good amount of consequences of MM, is compatible with a Π_1 -definition of NS_{ω_1} :

- if there is a supercompact cardinal which is a limit of supercompact cardinals, then in a forcing extension of V , SRP holds true and NS_{ω_1} is Π_1 -definable ([19]).

We shall give a negative answer to the question asked in the beginning of the chapter for the special case when $\kappa = \omega_1$ assuming BMM and the saturation of NS_{ω_1} by proving the following theorem.

²Here M_1 is the canonical inner model with one Woodin cardinal.

Theorem 1.1. *Assume BMM and that NS_{ω_1} is saturated. Then NS_{ω_1} is not boldface Π_1 -definable over H_{ω_2} .³*

In order to prove Theorem 1.1 we need to first modify the forcing designed by Asperó and Schindler as in [3] where they have shown that MM^{++} implies Axiom (*). Let $\varphi(A)$ be a Σ_1 statement and $A \subset H_{\omega_2}$. We show that if $\varphi(A)$ is consistent in certain way, then it can be forced by a stationary set preserving forcing — a version of Asperó–Schindler forcing that will be defined in Chapter 3. One aspect of the aforementioned consistency property roughly says that in some outer model of V there exists an N such that

- N is a stationary set preserving extension of H_{ω_2} , and
- $\varphi(A)$ holds true in N .

Unfortunately, the existence of such an N alone does not guarantee $\varphi(A)$ to be forcible by a stationary set preserving forcing, cf. [1, pp. 4] the remark on BMM° . The missing part is that

- N has to permit a well-founded elementary embedding of itself with ω_1 being the critical point.

This elementary embedding plays an important role in the Asperó–Schindler forcing framework. We say that $\varphi(A)$ is *weakly honestly consistent*⁴ if it has a witness N with the above properties, cf. Definition 2.11. We are then able to show that if a Σ_1 -formula is weakly honestly consistent, then it can be forced by a stationary set preserving forcing. This is the main task of Chapter 3. We state the theorem precisely now.

Theorem 1.2. *Assume that NS_{ω_1} is saturated and that $2^{\aleph_1} = \aleph_2$. Let φ be a Σ_1 -formula in $\mathcal{L}_{\dot{\epsilon}, \text{NS}_{\omega_1}}$ and let $A \subset H_{\omega_2}$ be a set in V . Suppose that $\varphi(A)$ is good⁵*

³We note that the assumption of this theorem is not a consequence of Axiom (*) as Axiom (*) does not imply that NS_{ω_1} is saturated; see [27, Section 6.1] for a detailed account of this matter.

⁴It is a weakening of “honest consistency”, cf. [1, Definition 1.8] and [24, Definition 2.8]. Another related concept is “1-honest consistency” where “1” stands for the closure under the $x \mapsto M_1^\#(x)$ operator, cf. [2, Definition 2.3]. $M_1^\#$ ’s conveniently give us a Woodin cardinal, however, we are not able to make them available under our assumptions. Both consistency properties were introduced by Asperó and Schindler.

⁵Formulas being “good” is a technical requirement so that we can apply Asperó–Schindler’s forcing method to formulas with parameters of size bigger than $\aleph_1$; in our case, the parameter is the nonstationary ideal itself. Cf. Definition 2.13.

and weakly honestly consistent. Then there is a stationary set preserving forcing $\mathbb{P}$ such that $V^{\mathbb{P}} \models \varphi(A)$.

The proof of this theorem is in essence the same as [3, Lemma 2.14] modulo minor modifications. First of all, our setting is simpler in the sense that no universally Baire sets is involved so that our semantic certificates only have two components: one is the “working” part which will add a term model witnessing the weak honest consistency of the given formula, the other is the “side condition” part which guarantees stationary set preservation. However in the original paper [3], the universally Baire set codes a set of $\mathbb{P}_{\max}$ conditions hence any model derived from the tree representing this universally Baire set is automatically well-founded. In our case, extra care needs to be taken in order to prove well-foundedness of the term model. The solution is, that in the definition of semantic certificates, we limit the ordinal height of the models which witness the weak honest consistency of $\varphi(A)$: at stage λ of the recursive definition of the forcing, the witness N has to be of ordinal height *strictly* less than λ . Indeed, we could as well prove the well-foundedness of the term model if we only required the ordinal height of N to be at most λ , but then we would not be able to show another nice and important property that club many forcings in the recursively defined sequence end-extend previous ones, cf. Lemma 3.22.

The restriction on the height of N prevents us from showing the canonical object derived from a filter forms a semantic certificate; we can only show that this canonical object is a semantic *quasi*-certificate which fortunately is sufficient to prove that our forcing preserves stationary sets. These details are handled in Chapter 3.

The assumption that NS_{ω_1} is saturated and that $2^{\aleph_1} = \aleph_2$ gives us a generic iteration $\mathcal{I}$ by the nonstationary ideal and its images of length $\omega_1 + 1$ with $H_{\omega_2}^V$ being the final iterate. Let N be a witness for the weakly honest consistency of $\varphi(A)$ and let $\sigma : N \rightarrow N^*$, where N^* is transitive, be an elementary embedding with $\text{crit}(\sigma) = \omega_1$. The iteration $\mathcal{I}$ will be further stretched by σ and a tail end of $\sigma(\mathcal{I})$ will be lifted up to the entire V . In the original paper [3], the existence of such an iteration whose final iterate is H_{ω_2} is again automatic by the definition of $\mathbb{P}_{\max}$.

Let us now turn our attention to the definability problem of NS_{ω_1} . Suppose that Theorem 1.1 is false, then there is a Σ_1 -formula φ_{stat} and an $X \subset \omega_1$ such

that for all $S \subset \omega_1$,

$$S \text{ is stationary} \iff \varphi_{\text{stat}}(S, X).$$

Roughly speaking, we will show that there is a two-step iteration of forcings which preserves stationary sets and adds a $D \subset \omega_1$ such that $\varphi_{\text{stat}}(D, X)$ holds but D is *nonstationary*. But such D must already exist in V by a typical application of BMM, a contradiction!

The following statement

$$\varphi(\text{NS}_{\omega_1}^{+V}) \equiv \exists D \subset \omega_1 (\varphi_{\text{stat}}(D, X) \wedge \forall S \in \text{NS}_{\omega_1}^{+V}, S \setminus D \text{ is stationary})$$

is shown to be good and weakly honestly consistent, cf. Definition 2.11. So that there is an Asperó–Schindler-style forcing $\mathbb{P}$, cf. Theorem 3.1, which is stationary set preserving and forces $\varphi(\text{NS}_{\omega_1}^{+V})$. This forcing $\mathbb{P}$ is exactly the first forcing of the two-step iteration as mentioned above. Let $D \subset \omega_1$ be stationary and co-stationary in $V^{\mathbb{P}}$ such that

$$\varphi_{\text{stat}}(D, X) \wedge \forall S \in \text{NS}_{\omega_1}^{+V}, S \setminus D \text{ is stationary}.$$

We now use the standard club-shooting forcing $\mathbb{S}$ to add a club in the complement of D so that D is nonstationary in $V^{\mathbb{P} * \mathbb{S}}$. By our assumption the formula $\varphi_{\text{stat}}(D, X)$ is Σ_1 so $\varphi_{\text{stat}}(D, X)$ still holds true in $V^{\mathbb{P} * \mathbb{S}}$. We have

$$V^{\mathbb{P} * \mathbb{S}} \models \exists D (\varphi_{\text{stat}}(D, X) \wedge D \text{ is nonstationary}).$$

The point here is that $\mathbb{P} * \mathbb{S}$ preserves stationary sets of V even though the stationarity of D is destroyed by $\mathbb{S}$. Applying BMM to the forcing $\mathbb{P} * \mathbb{S}$ we immediately have

$$V \models \exists D (\varphi_{\text{stat}}(D, X) \wedge D \text{ is nonstationary}),$$

but this contradicts the assumption that stationarity in V is defined by φ_{stat} . The technical details are handled in Chapter 4.

In the same chapter we also investigate descriptive set theoretic aspects of the definability problem of NS_{ω_1} . We first show that the presence of forcing axioms trivializes the usual Σ_n^1/Π_n^1 hierarchy in that if MA_{ω_1} holds then every subset of the generalized Baire space ${}^{\omega_1}\omega_1$ is Σ_1^1 . More generally, for any regular uncountable cardinal κ , if there is a $\lambda < \kappa$ such that $2^\lambda = 2^\kappa$, every subset of the space

$(^\kappa\kappa)^\ell$ is $\underline{\Sigma}_1^1$, where ℓ is a natural number, cf. Theorem 4.10. This means that NS_{ω_1} is always $\underline{\Pi}_1^1$ under MA_{ω_1} . We however introduce the so called $\text{uB-}\underline{\Sigma}_n^1/\underline{\Pi}_n^1$ hierarchy and show that under reasonable assumptions, NS_{ω_1} cannot be $\text{uB-}\underline{\Pi}_1^1$. This generalizes the nondefinability result of NS_{ω_1} . Specifically, we show that

Theorem 1.3. *Assume Γ^∞ -BMM and that there is a Woodin cardinal. Then NS_{ω_1} is not $\text{uB-}\underline{\Pi}_1^1$.*

An even stronger canonical property for NS_{ω_1} is the Σ_1 -completeness. We say that $A \subset {}^{\omega_1}2$ is Σ_1 -complete if for any Σ_1 -definable subset $B \subset {}^{\omega_1}2$ of H_{ω_2} , there is a continuous function $f : {}^{\omega_1}2 \rightarrow {}^{\omega_1}2$ such that for all $X \subset \omega_1$,

$$X \in B \iff f(X) \in A.$$

This property is stronger in the sense that Σ_1 -completeness directly implies that NS_{ω_1} is not Π_1 -nondefinable. Using the same argument as in the proof of [10, Theorem 49], one can show that NS_{ω_1} is indeed Σ_1 -complete in the constructible universe L . It was expected that strong forcing axioms, e.g. MM or MM^{++} , not only imply that NS_{ω_1} is Π_1 -nondefinable, but also Σ_1 -complete. However, a recent result of Lietz rules out such possibilities. Lietz's theorem says, assume MM , then NS_{ω_1} is *not* Σ_1 -complete, cf. [20, Corollary 2.6]. We sketch Lietz's proof.

Chapter 2

Preliminaries

We briefly review some relevant definitions and facts.

2.1 The saturation of NS_{ω_1}

Definition 2.1. Let κ be a regular uncountable cardinal and let I be an ideal on κ . We say that I is *precipitous* if every condition in the partial order $(I^+; \subset)$ forces that the generic ultrapower is well-founded.

Definition 2.2. Let κ be a regular uncountable cardinal and let I be an ideal on κ . We say that I is λ -*saturated* if the partial order $(I^+; \subset)$ has the λ -chain condition.

Let us make the following standard convention.

Definition 2.3. NS_{ω_1} is *saturated* if NS_{ω_1} is $\aleph_2$ -saturated.

Fact 2.4. Suppose that NS_{ω_1} is saturated. Then NS_{ω_1} is precipitous.

Proof. Cf. e.g., [17, Lemma 22.22]. □

Remark. Another way to see this fact is using the following characterizations:

- NS_{ω_1} is precipitous if and only if for all sufficiently large θ ,

$$\{X \prec H_\theta : X \text{ is countable and self-generic}\}$$

is projective stationary,⁶ and

⁶We say that $X \prec H_\theta$ is self-generic if the set $\{S \in \mathcal{P}(\omega_1) \cap X : X \cap \omega_1 \in S\}$ is X -generic for $(\text{NS}_{\omega_1}^+; \subset)$; and that $\mathcal{S} \in [H_\theta]^\omega$ is projective stationary if for all stationary $S \subset \omega_1$, the set $\{X \in \mathcal{S} : X \cap \omega_1 \in S\}$ is stationary.

- NS_{ω_1} is saturated if and only if for all sufficiently large θ ,

$$\{X \prec H_\theta : X \text{ is countable and self-generic}\}$$

contains a club.

These characterizations are due to Schindler and Woodin, respectively, cf. [7]. Then it is obvious that saturation of NS_{ω_1} implies precipitousness.

Definition 2.5. $\mathcal{I} = (M_i, g_i, \pi_{ij} : i < j \leq \omega_1)$ is a generic iteration by the nonstationary ideal (on ω_1) of M_0 and its images if

- (i) for each $i < \omega_1$, g_i is an M_i -generic filter for $(\text{NS}_{\omega_1}^{+M_i}; \subset)$ and $\pi_{i,i+1} : M_i \rightarrow M_{i+1}$ is the generic ultrapower embedding given by g_i ,
- (ii) for each limit ordinal $j \leq \omega_1$, M_j is the direct limit of $(M_i, \pi_{ik} : i < k < j)$, and $\pi_{ij} : M_i \rightarrow M_j$ is the induced embedding for each $i < j$.

We shall omit the g_i 's and simply write $\mathcal{I} = (M_i, \pi_{ij} : i < j \leq \omega_1)$ whenever the choice of generic filters is irrelevant. We call the models M_i 's iterates of $\mathcal{I}$.

Lemma 2.6 (folklore). *Assume that NS_{ω_1} is saturated and that $2^{\aleph_1} = \aleph_2$. Let N be a transitive model of ZFC^- in some outer model of V such that*

- (i) $\omega_1^N = \omega_1$,
- (ii) $H_{\omega_2}^V \in N$ and $N \models |\omega_2^V| = |H_{\omega_2}^V| = \aleph_1$, and
- (iii) $N \models \text{cf } \omega_2^V = \omega$.

Then in N there is a generic iteration $\mathcal{I} = (M_i, \pi_{ij} : i < j \leq \omega_1)$ by the nonstationary ideal of M_0 and its images such that $M_{\omega_1} = H_{\omega_2}^V$.

Proof. Fix a bijection $F : \omega_2^V \rightarrow H_{\omega_2}^V$ in V . Work in N . Define a sequence $(X_i : i \leq \omega_1)$ of sets by induction:

- $(X_0; \in, F \upharpoonright X_0) \prec (H_{\omega_2}^V; \in, F)$ is a countable elementary submodel such that the ordinals of X_0 are cofinal in ω_2^V ,
- $X_{i+1} = \{f(X_i \cap \omega_1) : f \text{ is a function in } X_i\}$, for $i < \omega_1$,

- $X_j = \bigcup \{X_i : i < j\}$, if $j \leq \omega_1$ is a limit ordinal.

We have a chain of elementary models:

$$(X_i; \in) \prec (X_j; \in) \prec (X_{\omega_1}; \in) \prec (H_{\omega_2}^V; \in), \quad i \leq j \leq \omega_1.$$

For $i \leq \omega_1$, let M_i be the transitive collapse of X_i and let $e_i : M_i \rightarrow X_i$ be the uncollapsing map. Let $\pi_{ij} = e_j^{-1} \circ e_i$ for $i \leq j \leq \omega_1$. Then $\pi_{i,i+1} : M_i \rightarrow M_{i+1}$ is a generic ultrapower embedding by $((\text{NS}_{\omega_1}^+)^{M_i}; \subset)$ since NS_{ω_1} is saturated, cf. [27, Lemma 3.12]. We remain to show that

$$M_{\omega_1} = H_{\omega_2}^V.$$

Claim 2.7. $\omega_2^V \subset X_{\omega_1}$.

Proof of Claim 2.7. It is easy to see $\omega_1 \subset X_{\omega_1}$ as we keep filling ordinals in the gap between $X_i \cap \omega_1$ and $X_{i+1} \cap \omega_1$ for ω_1 many times. Let $\{\alpha_n : n < \omega\} \subset X_{\omega_1}$ be a set of ordinals which is cofinal in ω_2^V . Since

$$X_{\omega_1} \models “\omega_1 \text{ is the largest cardinal}”,$$

there is a surjection $f_n : \omega_1 \rightarrow \alpha_n$ in X_{ω_1} for each $n < \omega$. This means

$$\alpha_n = f_n “\omega_1 \subset X_{\omega_1}”$$

for each $n < \omega$. Hence $\omega_2^V \subset X_{\omega_1}$. □ (Claim 2.7)

Claim 2.8. $(X_{\omega_1}; \in, F \restriction X_{\omega_1}) \prec (H_{\omega_2}^V; \in, F)$.

Proof of Claim 2.8. We prove this claim by induction. The case for X_0 is part of definition. Now suppose

$$(X_i; \in, F \restriction X_i) \prec (H_{\omega_2}^V; \in, F),$$

we want to show

$$(X_{i+1}; \in, F \restriction X_{i+1}) \prec (H_{\omega_2}^V; \in, F).$$

We only need to show that for all $A \in X_{i+1}$, there is $g \in X_i$ such that

$$F(g(X_i \cap \omega_1)) = A.$$

Fix $A \in X_{i+1}$ and let $f \in X_i$ be such that

$$A = f(X_i \cap \omega_1).$$

We may assume that $\text{dom}(f) = \omega_1$. As F is a bijection in V , we have

$$X_i \models \exists g : \omega_1 \rightarrow \omega_2^V \forall \xi < \omega_1 (F(g(\xi)) = f(\xi)).$$

Fix such $g \in X_i$. Then

$$X_i \models \forall \xi < \omega_1 (F(g(\xi)) = f(\xi)),$$

which means

$$H_{\omega_2}^V \models \forall \xi < \omega_1 (F(g(\xi)) = f(\xi)),$$

in particular,

$$F(g(X_i \cap \omega_1)) = f(X_i \cap \omega_1) = A. \quad \square \text{ (Claim 2.8)}$$

Recall that F is a bijection, we have

$$H_{\omega_2}^V = F''\omega_2^V \subset X_{\omega_1},$$

and hence $M_{\omega_1} = X_{\omega_1} = H_{\omega_2}^V$. $\square$ (Lemma 2.6)

Theorem 2.9 (Shoenfield absoluteness). *Let $M \subset N$ be transitive models of set theory such that $\omega_1^V \subset M$. Then*

$$H_{\omega_1}^M \prec_{\Sigma_1} H_{\omega_1}^N.$$

Shoenfield absoluteness is one of the key tools in the present thesis. Suppose $\psi(u, v)$ is a Σ_0 formula and $\exists x \psi(x, y)$ holds in some model, then by Shoenfield, in any model where y is hereditarily countable there is an x such that $\psi(x, y)$ holds.

2.2 Weak honest consistency

The concept of honest consistency was introduced by Schindler, cf. [24, Definition 2.8]. We use $\mathcal{L}_{\dot{\in}, \text{NS}_{\omega_1}}$ to denote the language of set theory $\mathcal{L}_{\dot{\in}}$ augmented with a unary predicate NS_{ω_1} for the nonstationary ideal on ω_1 , i.e. for any given model M and for any set $A \in M$,

$$(\text{NS}_{\omega_1}(A))^M \iff A \in \text{NS}_{\omega_1}^M.$$

Definition 2.10 (Schindler [24]). Let φ be a formula in $\mathcal{L}_{\dot{\in}, \text{NS}_{\omega_1}}$ and $A \in V$. Let $\theta = \aleph_1 + |\text{TC}(A)|$. We say that $\varphi(A)$ is *honestly consistent* if and only if for every function $F : H_{\omega_1} \rightarrow H_{\omega_1}$ which is strongly universally Baire in the codes, there is a set $N \in V^{\text{Col}(\omega, 2^\theta)}$ such that

- (i) N is a transitive model of ZFC^- ,
- (ii) $H_{\theta^+} \in N$ and $N \models \varphi(A)$,
- (iii) $\text{NS}_{\omega_1}^N \cap V = \text{NS}_{\omega_1}^V$, and
- (iv) N is F -closed.⁷

We define weak honest consistency by removing the mentioning of the function F and replacing “ N is F -closed” in the above definition with “ $\text{NS}_{\omega_1}^N$ is precipitous”. This weakened version is sufficient (and more convenient) for our purposes. It can be shown that honest consistency implies weak honest consistency under reasonably strong assumptions, e.g. Martin’s Maximum.

Definition 2.11. Let φ be a formula in $\mathcal{L}_{\dot{\in}, \text{NS}_{\omega_1}}$ and $A \subset H_{\omega_2}$ a set in V . We say that $\varphi(A)$ is *weakly honestly consistent* if and only if there is a set $N \in V^{\text{Col}(\omega, 2^{\aleph_1})}$ such that

- (i) N is a transitive model of ZFC^- ,
- (ii) $\{A, H_{\omega_2}\} \subset N$ and $N \models \varphi(A)$,
- (iii) $\text{NS}_{\omega_1}^N \cap V = \text{NS}_{\omega_1}^V$, and

⁷We shall not give definitions of “strongly universally Baire in the codes” and “ F -closed” as they are not used in the present thesis. They can be found in [24, Definitions 2.3 and 2.7].

(iv) $\text{NS}_{\omega_1}^N$ is precipitous.

Remark. We note that H_{ω_2} is a hidden parameter of $\varphi(A)$, which means that the weak honest consistency of $\varphi(A)$ is not automatically absolute across models with A as an element. We shall, however, suppress the occurrence of H_{ω_2} , namely, we always write $\varphi(A)$ instead of $\varphi(A, H_{\omega_2})$. The reader should have no problem filling in the correct hidden parameter, cf. e.g. the proof of Lemma 2.14. We sometimes also say $\varphi(A)$ is M -weakly honestly consistent if

$$M \models \text{“}\varphi(A) \text{ is weakly honestly consistent”},$$

cf. e.g. Lemma 3.8.

Remark. Also note that φ is a formula in the language $\mathcal{L}_{\dot{\epsilon}, \text{NS}_{\omega_1}^N}$. So $N \models \varphi(A)$ as in item (ii) really means $(N; \in, \text{NS}_{\omega_1}^N) \models \varphi(A)$. We shall mostly omit the predicate for the nonstationary ideal and just write $N \models \varphi(A)$ for simplicity.

We say that N is a *witness* for the weak honest consistency of $\varphi(A)$ if items (i)–(iv) as in Definition 2.11 are satisfied by N .

Lemma 2.12. *Let φ be a formula in $\mathcal{L}_{\dot{\epsilon}, \text{NS}_{\omega_1}^N}$ and $A \subset H_{\omega_2}$ a set in V . Suppose that $\varphi(A)$ is weakly honestly consistent. The following hold and are equivalent.*

- (1) *a countable witness for the weak honest consistency of $\varphi(A)$ exists in $V^{\text{Col}(\omega, 2^{\aleph_1})}$,*
- (2) *a countable witness for the weak honest consistency of $\varphi(A)$ exists in W for some outer model $W \supset V$ where $H_{\omega_2}^V$ is countable,*
- (3) *a countable witness for the weak honest consistency of $\varphi(A)$ exists in W for any outer model $W \supset V$ where $H_{\omega_2}^V$ is countable.*

Proof. The statement “there is a witness for the weak honest consistency of $\varphi(A)$ ” is Σ_1 in the parameters A and $H_{\omega_2}^V$ which are of size at most $|H_{\omega_2}^V| = 2^{\aleph_1}$. Then by Shoenfield absoluteness, the statement is absolute across all outer models of V where $H_{\omega_2}^V$ is countable and the witness can be chosen to be countable. The lemma follows. $\square$

We are interested in formulas which can be expressed in a “good” way.

Definition 2.13. Let φ be a formula in $\mathcal{L}_{\dot{\epsilon}, \aleph_{\omega_1}}$ and $A \subset H_{\omega_2}$ a set in V . We say that $\varphi(A)$ is *good* if and only if $\varphi(A)$ has the following form:

$$\varphi(A) \equiv \exists X \subset \omega_1 \forall Y \in A \psi(X, Y)$$

where ψ is a Σ_0 -formula in $\mathcal{L}_{\dot{\epsilon}, \aleph_{\omega_1}}$ and only uses parameters from H_{ω_2} .

Lemma 2.14. Let φ be a formula in $\mathcal{L}_{\dot{\epsilon}, \aleph_{\omega_1}}$ and $A \subset H_{\omega_2}$ a set in V . Suppose that $\varphi(A)$ is good and weakly honestly consistent. Then

$$\varphi(A) \wedge \text{cf } \omega_2^V = \omega$$

is weakly honestly consistent. Moreover, the witness N for the weak honest consistency of “ $\varphi(A) \wedge \text{cf } \omega_2^V = \omega$ ” can be chosen in a way such that

$$N \models |H_{\omega_2}^V| = \aleph_1.$$

Proof. Let θ be sufficiently large, and let $j : M \rightarrow H_\theta$ be an elementary embedding such that

- (a) M is transitive and has size $\aleph_1$,
- (b) $\text{crit}(j) = \omega_2^M > \omega_1$,
- (c) $\text{cf } \omega_2^M = \omega$, and
- (d) A and all the parameters of ψ are in the range of j ; we write $\bar{A}$ for the preimage of A , i.e. $j(\bar{A}) = A$.

As $\varphi(A)$ is weakly honestly consistent, let $N \in H_\theta^{\text{Col}(\omega, 2^{\aleph_1})}$ satisfy items (i)–(iv) as in Definition 2.11. We have $\{H_{\omega_2}^M, \bar{A}\} \in H_{\omega_2} \subset N$. It is easy to see $\bar{A} \subset A$ as the critical point of j is larger than ω_1 . Recall that $\varphi(A)$ is good, this means that

$$\varphi(\bar{A}) \equiv \exists X \subset \omega_1 \forall Y \in \bar{A} \psi(X, Y)$$

holds in N . The following are easily verified:

- (M.i) N is a transitive model of ZFC^- ,
- (M.ii) $\{\bar{A}, H_{\omega_2}^M\} \subset N$ and $N \models “\varphi(\bar{A}) \wedge \text{cf } \omega_2^M = \omega”$,⁸

⁸Here $H_{\omega_2}^M$ is the hidden parameter of $\varphi(\bar{A})$.

(M.iii) $\text{NS}_{\omega_1}^N \cap M = \text{NS}_{\omega_1}^M$,

(M.iv) $\text{NS}_{\omega_1}^N$ is precipitous, and

(M.v) $N \models |H_{\omega_2}^M| = \aleph_1$.

We have

$$H_{\theta}^{\text{Col}(\omega, 2^{\aleph_1})} \models \text{there is an } N \text{ that satisfies (M.i)–(M.v)}.$$

By Lemma 2.12,

$$M^{\text{Col}(\omega, (2^{\aleph_1})^M)} \models \text{there is an } N \text{ that satisfies (M.i)–(M.v)}.$$

Then by the elementarity of j ,

$$H_{\theta}^{\text{Col}(\omega, 2^{\aleph_1})} \models \text{there is an } N \text{ that satisfies (V.i)–(V.v)},$$

where items (V.i)–(V.v) are defined as following:

(V.i) N is a transitive model of ZFC^- ,

(V.ii) $\{A, H_{\omega_2}^V\} \subset N$ and $N \models “\varphi(A) \wedge \text{cf } \omega_2^V = \omega”$

(V.iii) $\text{NS}_{\omega_1}^N \cap V = \text{NS}_{\omega_1}^V$,

(V.iv) $\text{NS}_{\omega_1}^N$ is precipitous, and

(V.v) $N \models |H_{\omega_2}^V| = \aleph_1$.

The lemma is proved. □

The “ $\text{cf } \omega_2^V = \omega$ ” and “ $N \models |H_{\omega_2}^V| = \aleph_1$ ” parts in the above Lemma are used to show the existence of a generic iteration with H_{ω_2} being the final iterate.

Lemma 2.15. *Assume that NS_{ω_1} is saturated and $2^{\aleph_1} = \aleph_2$. Let φ be a formula in $\mathcal{L}_{\dot{\in}, \text{NS}_{\omega_1}}$ and $A \subset H_{\omega_2}$ a set in V . Suppose that $\varphi(A)$ is good and weakly honestly consistent. Then in $V^{\text{Col}(\omega, \omega_2)}$ there is a set N such that*

(i) N is a transitive model of ZFC^- ,

(ii) $\{A, H_{\omega_2}\} \subset N$ and $N \models \varphi(A)$,

(iii) $\text{NS}_{\omega_1}^N \cap V = \text{NS}_{\omega_1}^V$,

(iv) $\text{NS}_{\omega_1}^N$ is precipitous, and

(v) N has a generic iteration $\mathcal{I} = (M_i, \pi_{ij} : i < j \leq \omega_1)$ by the nonstationary ideal of M_0 and its images such that $M_{\omega_1} = H_{\omega_2}$.

Proof. By Lemma 2.14, let N be a witness for the weakly honest consistency of

$$\varphi(A) \wedge \text{cf } \omega_2^V = \omega$$

such that

$$N \models |H_{\omega_2}^V| = \aleph_1.$$

This means that items (i)–(iv) are satisfied by N . By Lemma 2.6, N has the generic iteration required by item (v). $\square$

2.3 Universally Baire sets and forcing axioms

Definition 2.16 (Feng–Magidor–Woodin [8]). Let $1 \leq k < \omega$. A set $A \subset \mathbb{R}^k$ is *universally Baire*⁹ if for all uncountable cardinal κ , there are ordinals α and β and trees $T \subset {}^{<\omega}(\omega^k \times \alpha)$ and $U \subset {}^{<\omega}(\omega^k \times \beta)$ such that $A = p[T]$ and for all forcing $\mathbb{P} \in H_\kappa$,

$$\Vdash_{\mathbb{P}} p[T] = \mathbb{R}^k \setminus p[U].$$

Let $\mathbb{P}$ be any forcing and let $g \subset \mathbb{P}$ be a generic filter. Let A be a universally Baire set of reals (or tuples of reals) and let trees T, U witness the universal Baireness of A . Then there is a canonical new version of A in $V[g]$, for which we write

$$A_g = p[T]^{V[g]}.$$

If the generic filter g is clear from the context, we shall write A^* instead of A_g . We sometimes work with models which are not generic extensions of V . Say N is

⁹The original definition of universally Baire sets is given in terms of topological spaces and continuous functions, cf. [8]. Definition 2.16 is an equivalent formulation stated as a theorem that is proved in the same paper. It is sometimes more convenient to use this definition instead of the original topological one.

an extension of an initial segment of V which exists in some outer model of V and N has T, U as elements, then we also use A^* to denote $p[T]^N$.

Universal Baire-ness is a central concept in modern set theory, however, not much can be said about universally Baire sets without assuming large cardinals. The standard assumption is the existence of proper class many Woodin cardinals. We write Γ^∞ for the class of all universally Baire sets.

Theorem 2.17 (Woodin). *Assume that there is a proper class of Woodin cardinals. The following hold:*

- (i) Γ^∞ is closed under projections and complements,
- (ii) for every universally Baire $A \subset \mathbb{R}$, $L(A, \mathbb{R}) \models \text{AD}^+$.

Proof. For item (i), see e.g. [3, Theorem 2.10]. Item (ii) follows from the arguments in [26]. $\square$

Let us turn our attention to forcing axioms. We shall mainly focus on Martin's Maximum and its variations, and the $\mathbb{P}_{\max}$ -axiom (*).

Definition 2.18 (Foreman–Magidor–Shelah [9]). (1) *Martin's Maximum*, abbr.

MM, is the statement that for all stationary set preserving forcing $\mathbb{P}$ and for all $\mathcal{D}$ of $\aleph_1$ -many dense subsets of $\mathbb{P}$, there exists a filter $g \subset \mathbb{P}$ which is $\mathcal{D}$ -generic.

- (2) *Martin's Maximum⁺⁺*, abbr. MM^{++} , is the statement that for all stationary set preserving forcing $\mathbb{P}$, for all $\mathcal{D}$ of $\aleph_1$ -many dense subsets of $\mathbb{P}$, and for all $\{\tau_i : i < \omega_1\}$ of $\mathbb{P}$ -names for stationary subsets of ω_1 , there exists a filter $g \subset \mathbb{P}$ which is $\mathcal{D}$ -generic such that

$$\tau_i^g = \{\xi < \omega_1 : \exists p \in g (p \Vdash \xi \in \tau)\}$$

is stationary for every $i < \omega_1$.

Lemma 2.19 (folklore). *The following are equivalent:*

- (1) MM,
- (2) for any stationary set preserving forcing $\mathbb{P}$, for any sufficiently large θ , and for any $X \subset H_\theta$ of size $\aleph_1$, there is an elementary embedding $j : M \rightarrow H_\theta$ such that the following hold:

- (i) M is transitive and has size $\aleph_1$,
- (ii) $\omega_1 \cup X \cup \{\mathbb{P}\} \subset \text{ran}(j)$, and
- (iii) there is an M -generic filter for $j^{-1}(\mathbb{P})$.

Proof. See e.g. [6, Theorem 1.3]. □

Definition 2.20 (Goldstern–Shelah [13]). (1) *Bounded Martin’s Maximum*, abbr.

BMM, is the statement that for all stationary set preserving forcing $\mathbb{P}$ and for all $\mathcal{D}$ of $\aleph_1$ -many dense subsets of $\mathbb{P}$ such that every set in $\mathcal{D}$ has size at most $\aleph_1$, there exists a filter $g \subset \mathbb{P}$ which is $\mathcal{D}$ -generic.

- (2) *Bounded Martin’s Maximum⁺⁺*, abbr. BMM^{++} , is the statement that for all stationary set preserving forcing $\mathbb{P}$, for all $\mathcal{D}$ of $\aleph_1$ -many dense subsets of $\mathbb{P}$ such that every set in $\mathcal{D}$ has size at most $\aleph_1$, and for all $\{\tau_i : i < \omega_1\}$ of $\mathbb{P}$ -names for stationary subsets of ω_1 , there exists a filter $g \subset \mathbb{P}$ which is $\mathcal{D}$ -generic such that

$$\tau_i^g = \{\xi < \omega_1 : \exists p \in g (p \Vdash \xi \in \tau)\}$$

is stationary for every $i < \omega_1$.

Theorem 2.21 (Bagaria [4]). *The following are equivalent:*

- (1) BMM,
- (2) for all stationary set preserving forcing $\mathbb{P}$,

$$H_{\omega_2} \prec_{\Sigma_1} H_{\omega_2}^{V^{\mathbb{P}}}.$$

Theorem 2.22 (Bagaria [4]). *The following are equivalent:*

- (1) BMM^{++} ,
- (2) for all stationary set preserving forcing $\mathbb{P}$,

$$(H_{\omega_2}; \in, \text{NS}_{\omega_1}) \prec_{\Sigma_1} (H_{\omega_2}^{V^{\mathbb{P}}}; \in, \text{NS}_{\omega_1}^{V^{\mathbb{P}}}).$$

Γ^∞ -Bounded Martin's Maximum and its “++” variation were also introduced by Woodin [27, Definitions 10.122 and 10.123]. We give equivalent reformulations using generic absoluteness.

Definition 2.23 (Woodin [27]). (1) Γ^∞ -Bounded Martin's Maximum, abbr. Γ^∞ -BMM, is the statement that for all stationary set preserving forcing $\mathbb{P}$ and for all universally Baire set $A \subset \mathbb{R}$,

$$(H_{\omega_2}; \in, A) \prec_{\Sigma_1} (H_{\omega_2}^{V^\mathbb{P}}; \in, A^*).$$

(2) Γ^∞ -Bounded Martin's Maximum $^{++}$, abbr. Γ^∞ -BMM $^{++}$, is the statement that for all stationary set preserving forcing $\mathbb{P}$ and for all universally Baire set $A \subset \mathbb{R}$,

$$(H_{\omega_2}; \in, \text{NS}_{\omega_1}, A) \prec_{\Sigma_1} (H_{\omega_2}^{V^\mathbb{P}}; \in, \text{NS}_{\omega_1}^{V^\mathbb{P}}, A^*).$$

Definition 2.24 (Woodin [27]). Axiom $(*)$ is the conjunction of following statements:

- (i) AD holds in $L(\mathbb{R})$,
- (ii) there exists a filter $g \subset \mathbb{P}_{\max}$ which is generic over $L(\mathbb{R})$ such that $\mathcal{P}(\omega_1) \subset L(\mathbb{R})[g]$.

A strengthening of Axiom $(*)$ was introduced by Schindler [24].

Definition 2.25 (Schindler [24]). Let $\Gamma \supset \mathcal{P}(\mathbb{R}) \cap L(\mathbb{R})$ be a pointclass. $(*)_\Gamma$ is the conjunction of the following statements:

- (i) every set in Γ is determined,
- (ii) there exists a filter $g \subset \mathbb{P}_{\max}$ which is Γ -generic (and hence generic over $L(\mathbb{R})$) such that $\mathcal{P}(\omega_1) \subset L(\mathbb{R})[g]$.¹⁰

Theorem 2.26 (Asperó–Schindler [3]). *Assume that there is a proper class of Woodin cardinals. Then MM^{++} implies $(*)_{\Gamma^\infty}$.*

Theorem 2.27 (Asperó–Schindler [3]). *Assume that there is a proper class of Woodin cardinals. The following are equivalent:*

¹⁰Here Γ -generic should be understood as $\tilde{\Gamma}$ -generic, where $\tilde{\Gamma}$ is the collection of subsets of $\mathbb{P}_{\max}$ that are coded by sets in Γ .

- (1) $(*)_{\Gamma^\infty}$,
- (2) Γ^∞ -BMM $^{++}$.

2.4 The stationary tower forcing

The stationary tower forcing was designed by Woodin. We refer the reader to Larson's book [18] for a comprehensive account of the stationary tower forcing. We only give definitions and facts that are relevant to the present thesis.

Definition 2.28. Let $X \neq \emptyset$ and let $C \subset \mathcal{P}(X)$. We say that C is a *club* in $\mathcal{P}(X)$ if there is a function $F : [X]^{<\omega} \rightarrow X$ such that

$$C = \{Z : F[[Z]^{<\omega}] \subset Z\}.$$

Definition 2.29. Let S be a set such that $\cup S \neq \emptyset$. We say that S is *stationary* if $C \cap S \neq \emptyset$ for every C which is club in $\mathcal{P}(\cup S)$.

Definition 2.30. Let δ be an inaccessible cardinal. Then the *(full) stationary tower (up to δ)* is

$$\mathbb{P}_{<\delta} = \{S \in H_\delta : S \text{ is stationary}\},$$

and for $S, T \in \mathbb{P}_{<\delta}$, $S \leq T$ if $\cup S \supset \cup T$ and $Z \cap (\cup T) \in T$ for every $Z \in S$.

Definition 2.31. Let δ be an inaccessible cardinal. Then the *countable tower (up to δ)* is

$$\mathbb{Q}_{<\delta} = \{S \in H_\delta : S \text{ is stationary and } \forall Z \in S, Z \text{ is countable}\}$$

with the order induced from $\mathbb{P}_{<\delta}$.

Fact 2.32 (Woodin). *Let δ be a Woodin cardinal. Let $\kappa < \delta$ be an infinite cardinal and let α be an ordinal below δ . Then there is $S \in \mathbb{P}_{<\delta}$ such that for any filter $g \ni S$ which is $\mathbb{P}_{<\delta}$ -generic over V , there is an elementary embedding $j : V \rightarrow M$ in $V[g]$ such that*

- (i) M is transitive,
- (ii) ${}^{<\delta}M \cap V[g] \subset M$,

(iii) $\text{crit}(j) = \kappa^+$ and $j(\kappa^+) > \alpha$,

(iv) for any $T \in \mathbb{P}_{<\delta}$, $T \in g$ if and only if $j[\cup T] \in j(T)$.¹¹

Fact 2.33 (Woodin). *Let δ be a Woodin cardinal. Let $g \subset \mathbb{Q}_{<\delta}$ be generic over V . Then there is an elementary embedding $j : V \rightarrow M$ in $V[g]$ such that*

(i) M is transitive,

(ii) ${}^{<\delta}M \cap V[g] \subset M$,

(iii) $\text{crit}(j) = \omega_1$ and $j(\omega_1) = \delta$,

(iv) for any $T \in \mathbb{P}_{<\delta}$, $T \in g$ if and only if $j[\cup T] \in j(T)$.

¹¹We may let, e.g., $S = \{X \prec H_\theta : |X| = \kappa \wedge \kappa \subset X\}$, where $\theta > \alpha$ is an inaccessible cardinal.

Chapter 3

A version of Asperó–Schindler forcing

We prove the following theorem in this chapter.

Theorem 3.1. *Assume that NS_{ω_1} is saturated and that $2^{\aleph_1} = \aleph_2$. Let φ be a formula in $\mathcal{L}_{\dot{\in}, \text{NS}_{\omega_1}}$ and let $A \subset H_{\omega_2}$ be a set in V . Suppose that $\varphi(A)$ is good (cf. Definition 2.13) and weakly honestly consistent (cf. Definition 2.11). Then there is a stationary set preserving forcing $\mathbb{P}$ such that*

$$V^{\mathbb{P}} \models \varphi(A).$$

While the saturation of NS_{ω_1} and $2^{\aleph_1} = \aleph_2$ are the only assumptions of Theorem 3.1, a few other key tools are needed in order to prove this theorem. The main task of Section 3.1 is to enable us to make other assumptions (cf. Convention 3.9). Then in Section 3.2 we define the forcing $\mathbb{P}$ under the new assumptions. Section 3.3 provides a method to “read off” a certificate from a filter for $\mathbb{P}$ which will be useful in the proof of the stationary set preservation of $\mathbb{P}$. Finally, we prove $\mathbb{P}$ preserves stationary subsets of ω_1 and verify $V^{\mathbb{P}} \models \varphi(A)$ in Section 3.4.

For the purposes of this chapter, we will need slightly stronger versions of Lemmas 2.14 and 2.12.

Lemma 3.2. *Let φ be a formula in $\mathcal{L}_{\dot{\in}, \text{NS}_{\omega_1}}$ and let $A \subset H_{\omega_2}$ be a set in V . Suppose that $\varphi(A)$ is good and weakly honestly consistent. Then there is an ordinal $\eta_A < (2^{\aleph_1})^{+V}$ such that in $V^{\text{Col}(\omega, 2^{\aleph_1})}$, there is an N satisfying*

- (i) *N witnesses the weak honest consistency of “ $\varphi(A) \wedge \text{cf } \omega_2^V = \omega$ ” and $H_{\omega_2}^V$ has size $\aleph_1$ in N , and*

(ii) the ordinal height of N is η_A , i.e. $\text{Ord}^N = \eta_A$.¹²

Proof. By Lemma 2.14, let $N \in V^{\text{Col}(\omega, 2^{\aleph_1})}$ be such that item (i) holds; and by Shoenfield absoluteness, we may assume that N is countable in $V^{\text{Col}(\omega, 2^{\aleph_1})}$. Now we simply let $\eta_A = \text{Ord}^N$. Then $\eta_A < (2^{\aleph_1})^{+V}$ because

$$(2^{\aleph_1})^{+V} = \omega_1^{V^{\text{Col}(\omega, 2^{\aleph_1})}}$$

and N is countable in $V^{\text{Col}(\omega, 2^{\aleph_1})}$. □

Definition 3.3. For the rest of the present chapter, we use η_A to denote a fixed ordinal below $(2^{\aleph_1})^{+V}$ such that Lemma 3.2 holds.

Lemma 3.4. Let φ be a formula in $\mathcal{L}_{\dot{\epsilon}, \text{NS}_{\omega_1}}$ and let $A \subset H_{\omega_2}$ be a set in V . Suppose that $\varphi(A)$ is good and weakly honestly consistent. The following hold and are equivalent.

- (1) a countable N satisfying items (i) and (ii) as in Lemma 3.2 exists in $V^{\text{Col}(\omega, 2^{\aleph_1})}$,
- (2) a countable N satisfying items (i) and (ii) as in Lemma 3.2 exists in W for some outer model $W \supset V$ where $H_{\omega_2}^V$ is countable,
- (3) a countable N satisfying items (i) and (ii) as in Lemma 3.2 exists in W for any outer model $W \supset V$ where $H_{\omega_2}^V$ is countable.

Proof. The statement “there is an N satisfying items (i) and (ii) as in Lemma 3.2” is Σ_1 in the parameters η_A , A , and $H_{\omega_2}^V$ which are of size at most $|H_{\omega_2}^V| = 2^{\aleph_1}$. Then (1) holds by Lemma 3.2; and by Shoenfield absoluteness, (2) and (3) hold and are equivalent to (1). □

3.1 Preparation

From now on until the end of the present chapter, let us assume that NS_{ω_1} is saturated and $2^{\aleph_1} = \aleph_2$, let us also write $\kappa = \omega_3^V$ and $\omega_1 = \omega_1^V$ for the sake of notational simplicity. Let d be a filter for $\text{Col}(\kappa, \kappa)$ which is generic over V .

¹²It is trivial to “prove” the existence of η_A . This η_A was used to show the well-foundedness of the “term model” produced in Section 3.3 in earlier versions of the thesis; in the present version, it is only used in Lemmas 3.4, 3.6, and 3.26 in a non-essential way. The author feels that it does not hurt to keep this triviality.

Lemma 3.5. *The following hold.*

- (i) *There is a $\diamond_\kappa$ -sequence in $V[d]$,*
- (ii) *$2^{<\kappa} = \kappa$ in $V[d]$,*
- (iii) *$H_\kappa^V = H_\kappa^{V[d]}$,*
- (iv) *$H_{\omega_2}^V = H_{\omega_2}^{V[d]}$,*
- (v) *$\text{NS}_{\omega_1}^V = \text{NS}_{\omega_1}^{V[d]}$, and*
- (vi) *NS_{ω_1} is saturated and $2^{\aleph_1} = \aleph_2$ in $V[d]$.*

Proof. For items (i) and (ii), cf. e.g. [23, Theorem 6.46] and [23, Lemma 6.42]. Item (iii) holds because $\text{Col}(\kappa, \kappa)$ is $<\kappa$ -closed. Items (iv)–(vi) are implied by (iii). $\square$

Let $(\bar{A}_\lambda : \lambda < \kappa)$ be a $\diamond_\kappa$ -sequence in $V[d]$. Let $c : \kappa \rightarrow H_\kappa$ be a bijection in $V[d]$. For $\lambda < \kappa$, let $Q_\lambda = c''\lambda$ and $A_\lambda = c''\bar{A}_\lambda$.

Lemma 3.6. *There is a club $C \subset \kappa$ in $V[d]$ such that the following hold for all $\lambda \in C$.*

- (Q.1) $\{\eta_A, A, H_{\omega_2}\} \subset Q_\lambda$,
- (Q.2) Q_λ is transitive,
- (Q.3) $Q_\lambda \cap \text{Ord} = \lambda$, and
- (Q.4) $(Q_\lambda; \in) \prec (H_\kappa; \in)$.

Proof. We only need to show that conditions (Q.1) through (Q.4) hold for unboundedly many λ 's as these conditions hold automatically at limit stages. It is easy to see (Q.1) holds for all sufficiently large λ 's. If $(\lambda_n : n < \omega)$ is a sequence of ordinals below κ such that the transitive closure of Q_{λ_n} is contained in Q_{λ_m} whenever $n < m < \omega$, then (Q.2) is satisfied by λ where $\lambda = \sup\{\lambda_n : n < \omega\}$. If $(\lambda_n : n < \omega)$ is a sequence of ordinals below κ such that Q_{λ_n} is transitive for each $n < \omega$ and

$$\max\{\text{Ord}^{Q_{\lambda_n}}, \lambda_n\} < \text{Ord}^{Q_{\lambda_m}}$$

whenever $n < m < \omega$, then (Q.3) is satisfied by λ where $\lambda = \sup\{\lambda_n : n < \omega\}$. As for (Q.4), if $(\lambda_n : n < \omega)$ is a sequence of ordinals below κ such that the closure of Q_{λ_n} under certain fixed set of Skolem functions is contained in Q_{λ_m} whenever $n < m < \omega$, then Q_λ is closed under the same set of Skolem functions where $\lambda = \sup\{\lambda_n : n < \omega\}$. The lemma now follows. $\square$

Let us fix a club $C \subset \kappa$ in $V[d]$ such that conditions (Q.1) through (Q.4) as in the above lemma are satisfied for all $\lambda \in C$.

Lemma 3.7. *The principle $(\diamond)$ holds true in $V[d]$, where $(\diamond)$ is the statement that for all $P, B \subset H_\kappa$, the set*

$$\{\lambda \in C : (Q_\lambda; \in, P \cap Q_\lambda, A_\lambda) \prec (H_\kappa; \in, P, B)\}$$

is stationary.

Proof. The argument showing (Q.4) in Lemma 3.6 also gives us that in $V[d]$, for all $P, B \subset H_\kappa$, the set

$$\{\lambda \in C : (Q_\lambda; \in, P \cap Q_\lambda, B \cap Q_\lambda) \prec (H_\kappa; \in, P, B)\}$$

contains a club. Since $(\bar{A}_\lambda : \lambda < \kappa)$ is a $\diamond_\kappa$ -sequence and c is a bijection, the set

$$\{\lambda \in C : B \cap Q_\lambda = A_\lambda\}$$

is stationary. We obtain $(\diamond)$ by taking the intersection of these two sets. $\square$

Lemma 3.8. $V[d] \models \text{“}\varphi(A) \text{ is weakly honestly consistent”}$.

Cf. the remark below Definition 2.11.

Proof of Lemma 3.8. By Shoenfield absoluteness, let $N \in V[d]^{\text{Col}(\omega, H_{\omega_2}^V)}$ witness the V -weak honest consistency of $\varphi(A)$. By Lemma 3.5 items (iv) and (v), N also witnesses the $V[d]$ -weak honest consistency of $\varphi(A)$. $\square$

As $\text{Col}(\kappa, \kappa)$ is stationary set preserving, Theorem 3.1 is proved if we can find a stationary set preserving forcing $\mathbb{P}$ in $V[d]$ such that whenever $h \subset \mathbb{P}$ is generic over $V[d]$,

$$V[d * h] \models \varphi(A).$$

For the ease of notations, we shall make the following assumptions as listed in Convention 3.9 about our universe V , and produce the desired forcing $\mathbb{P}$ in such V .

Convention 3.9. For the rest of the present chapter, we shall assume the following hold in V .

- (i) NS_{ω_1} is saturated,
- (ii) $\kappa = (2^{\aleph_1})^+$, and $(\bar{A}_\lambda : \lambda < \kappa)$ is a $\diamond_\kappa$ -sequence,
- (iii) $c : \kappa \rightarrow H_\kappa$ is a bijection, and for $\lambda < \kappa$, $Q_\lambda = c''\lambda$ and $A_\lambda = c''\bar{A}_\lambda$,
- (iv) $C \subset \kappa$ is a club such that conditions (Q.1)–(Q.4) as in Lemma 3.6 hold for all $\lambda \in C$, and
- (v) the principle $(\diamond)$ holds (cf. Lemma 3.7).

3.2 The definition of the forcing

We shall recursively define a sequence $(\mathbb{P}_\lambda : \lambda \in C \cup \{\kappa\})$ of forcings and the forcing $\mathbb{P}$ as in Theorem 3.1 will be $\mathbb{P}_\kappa$. Let $\lambda \in C \cup \{\kappa\}$ and suppose that $(\mathbb{P}_\mu : \mu \in C \cap \lambda)$ has already been defined. We aim to define $\mathbb{P}_\lambda$.

Definition 3.10. An object

$$\mathfrak{C} = (N, (\lambda_\delta, X_\delta : \delta \in K))$$

is a λ -semantic certificate (relative to $A, H_{\omega_2}, (A_\mu : \mu \in C \cap \lambda), (Q_\mu : \mu \in C \cap \lambda)$ and $(\mathbb{P}_\mu : \mu \in C \cap \lambda)$) if and only if the following conditions are satisfied:

- (C.1) N witnesses the weak honest consistency of “ $\varphi(A) \wedge \text{cf } \omega_2^V = \omega$ ” and $H_{\omega_2}^V$ has size $\aleph_1$ in N (cf. Definition 2.11 and Lemma 2.14),
- (C.2) the ordinal height of N is less than λ , i.e. $\text{Ord}^N < \lambda$,
- (C.3) $K \subset \omega_1$,
- (C.4) for all $\delta \in K$, $\lambda_\delta \in C \cap \lambda$, and if $\gamma < \delta$ is in K , then $\lambda_\gamma < \lambda_\delta$ and $X_\gamma \cup \{\lambda_\gamma\} \subset X_\delta$, and

($\mathfrak{C}.5$) for all $\delta \in K$, $X_\delta \prec (Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta})$ and $X_\delta \cap \omega_1 = \delta$.

Definition 3.11. An object

$$\mathfrak{C} = (N, (\lambda_\delta, X_\delta : \delta \in K))$$

is a λ -semantic quasi-certificate (relative to $A, H_{\omega_2}, (A_\mu : \mu \in C \cap \lambda), (Q_\mu : \mu \in C \cap \lambda)$ and $(\mathbb{P}_\mu : \mu \in C \cap \lambda)$) if and only if conditions ($\mathfrak{C}.1$) and ($\mathfrak{C}.3$)–($\mathfrak{C}.5$) as in Definition 3.10 are satisfied by $\mathfrak{C}$.

Remark. We shall omit the prefix “ λ –” and simply call such $\mathfrak{C}$ as in the above two definitions a *semantic certificate* or *semantic quasi-certificate* when the context causes no confusion. In fact, such differentiation is only needed in Lemma 3.22.

Remark. The reader who is already familiar with the original “MM⁺⁺ implies (*)” paper [3] by Asperó and Schindler should not confuse our “semantic quasi-certificate” with their “potential semantic certificate” as they are not the same thing. In the case of [3], an object being a potential semantic certificate was to stress that it only becomes a semantic certificate if it can be “certified” by a collection of $\mathcal{L}$ -formulas satisfying their condition ($\Sigma.8$) among other things, cf. [3, Definition 3.3]. We introduce the so called semantic quasi-certificate because we can only prove a weaker version of [3, Lemma 3.7], cf. Lemma 3.27. Or in other words, when we try to derive a certificate-like object from a filter, we can only guarantee conditions ($\mathfrak{C}.1$) and ($\mathfrak{C}.3$)–($\mathfrak{C}.5$) to be satisfied, but we cannot make sure that the ordinal height of the witness N is strictly less than λ . However, it is still worth noting that even in our case, an object being a semantic certificate alone does not automatically make it part of the forcing. We avoid excessive notations by not defining what is a potential semantic certificate.

Definition 3.12. The language $\mathcal{L}$ has the following constants:

$\underline{x}$ for each $x \in H_\kappa$	intended to denote x itself,
$\dot{\alpha}$ for each $\alpha < \omega_1$	as terms for elements of N ,
$\dot{N}$	intended to denote N ,
$\dot{N}S_{\omega_1}$	intended to denote N ’s predicate for the nonstationary ideal, and
$\dot{X}_\delta$ for each $\delta < \omega_1$	intended to denote X_δ .

The formulas of $\mathcal{L}$ are expressions of the following forms:

$$\begin{aligned}
 \ulcorner \dot{N} \models \Phi(\dot{\alpha}_1, \dots, \dot{\alpha}_k, \underline{x}_1, \dots, \underline{x}_\ell) \urcorner & \quad \text{for } \Phi \text{ a formula in } \mathcal{L}_{\dot{\in}, \text{NS}_{\omega_1}}, \\
 & \alpha_1, \dots, \alpha_k < \omega_1, \text{ and} \\
 & x_1, \dots, x_\ell \in H_{\omega_2}^V \cup \{\omega_2^V, H_{\omega_2}^V, A\}, \\
 \ulcorner \dot{\alpha} = \underline{\xi} \urcorner & \quad \text{for } \alpha < \omega_1, \xi < \kappa, \\
 \ulcorner \underline{\delta} \mapsto \underline{\mu} \urcorner & \quad \text{for } \delta < \omega_1, \mu < \kappa, \text{ and} \\
 \ulcorner \underline{x} \in \dot{X}_\delta \urcorner & \quad \text{for } \delta < \omega_1, x \in H_\kappa.
 \end{aligned}$$

$\mathcal{L}^\lambda$ -constants and $\mathcal{L}^\lambda$ -formulas are, respectively, $\mathcal{L}$ -constants and $\mathcal{L}$ -formulas that have no mentioning of elements outside Q_λ .

Let us fix a coding method for the language $\mathcal{L}$.¹³

Definition 3.13. Let $\#(\Phi)$ denote the Gödel number of Φ . $\mathcal{L}$ -constants and $\mathcal{L}$ -formulas are coded as:

$$\begin{aligned}
 \underline{x} &= (0, x) \quad \text{for } x \in H_\kappa, \\
 \dot{\alpha} &= (1, \alpha) \quad \text{for } \alpha < \omega_1, \\
 \dot{X}_\delta &= (2, \delta) \quad \text{for } \delta < \omega_1, \\
 \ulcorner \dot{N} \models \Phi(\dot{\alpha}_1, \dots, \dot{\alpha}_k, \underline{x}_1, \dots, \underline{x}_\ell) \urcorner &= (3, \#(\Phi), (1, \alpha_1), \dots, (1, \alpha_k), (0, x_1), \dots, (0, x_\ell)) \\
 & \quad \text{for } \Phi \text{ a formula in } \mathcal{L}_{\dot{\in}, \text{NS}_{\omega_1}}, \\
 & \alpha_1, \dots, \alpha_k < \omega_1, \text{ and} \\
 & x_1, \dots, x_\ell \in H_{\omega_2}^V \cup \{\omega_2^V, H_{\omega_2}^V, A\}, \\
 \ulcorner \dot{\alpha} = \underline{\xi} \urcorner &= (4, (1, \alpha), (0, \xi)) \quad \text{for } \alpha < \omega_1, \xi < \kappa, \\
 \ulcorner \underline{\delta} \mapsto \underline{\mu} \urcorner &= (5, (0, \delta), (0, \mu)) \quad \text{for } \delta < \omega_1, \mu < \kappa, \text{ and} \\
 \ulcorner \underline{x} \in \dot{X}_\delta \urcorner &= (6, (0, x), (2, \delta)) \quad \text{for } \delta < \omega_1, x \in H_\kappa.
 \end{aligned}$$

Definition 3.14. A collection Σ of $\mathcal{L}^\lambda$ -formulas is an $\mathcal{L}^\lambda$ -syntactic certificate if and only if there is a λ -semantic certificate

$$\mathfrak{C} = (N, (\lambda_\delta, X_\delta : \delta \in K))$$

¹³The exact way of coding is not important. We only need to guarantee that Lemma 3.22 holds true.

and a surjection $e : \omega_1 \rightarrow N$ such that the following conditions are satisfied:

$$(\Sigma.1) \quad \ulcorner \dot{N} \models \Phi(\dot{\alpha}_1, \dots, \dot{\alpha}_k, \underline{x}_1, \dots, \underline{x}_\ell) \urcorner \in \Sigma \text{ iff}$$

- (i) $\alpha_1, \dots, \alpha_k < \omega_1$,
- (ii) $x_1, \dots, x_\ell \in H_{\omega_2}^V \cup \{\omega_2^V, H_{\omega_2}^V, A\}$, and
- (iii) $N \models \Phi(e(\alpha_1), \dots, e(\alpha_k), x_1, \dots, x_\ell)$,

$$(\Sigma.2) \quad \ulcorner \underline{\delta} \mapsto \underline{\mu} \urcorner \in \Sigma \text{ iff } \delta \in K \text{ and } \mu = \lambda_\delta,$$

$$(\Sigma.3) \quad \ulcorner \underline{x} \in \dot{X}_\delta \urcorner \in \Sigma \text{ iff } \delta \in K \text{ and } x \in X_\delta, \text{ and}$$

$$(\Sigma.4) \quad \text{if } \delta \in K, \text{ then } [\Sigma]^{<\omega} \cap X_\delta \cap E \neq \emptyset \text{ for every } E \subset \mathbb{P}_{\lambda_\delta} \text{ which is dense in } \mathbb{P}_{\lambda_\delta} \\ \text{and definable over the structure } (Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta}) \text{ using parameters in } X_\delta.$$

Definition 3.15. A collection Σ of $\mathcal{L}^\lambda$ -formulas is an $\mathcal{L}^\lambda$ -*syntactic quasi-certificate* if and only if there is a λ -semantic quasi-certificate

$$\mathfrak{C} = (N, (\lambda_\delta, X_\delta : \delta \in K))$$

and a surjection $e : \omega_1 \rightarrow N$ such that conditions $(\Sigma.1)$ – $(\Sigma.4)$ as in Definition 3.14 are satisfied by Σ , $\mathfrak{C}$ and e .

Remark. We shall omit the prefix “ $\mathcal{L}^\lambda$ –” and simply call such Σ as in the above two definitions a *syntactic certificate* or *syntactic quasi-certificate* when the context causes no confusion. In fact, such differentiation is only needed in Lemma 3.22.

Lemma 3.16. *Let Σ be a syntactic quasi-certificate. Then there is a unique semantic quasi-certificate*

$$\mathfrak{C} = (N, (\lambda_\delta, X_\delta : \delta \in K))$$

and a unique surjection

$$e : \omega_1 \rightarrow N$$

such that conditions $(\Sigma.1)$, $(\Sigma.2)$ and $(\Sigma.3)$ as in Definition 3.14 are satisfied.

Proof. Suppose

$$\mathfrak{C} = (N, (\lambda_\delta, X_\delta : \delta \in K)), \quad e : \omega_1 \rightarrow N$$

and

$$\mathfrak{C}' = (N', (\lambda'_\delta, X'_\delta : \delta \in K')), \quad e' : \omega_1 \rightarrow N'$$

are semantic quasi-certificates and surjections that both satisfy conditions $(\Sigma.1)$, $(\Sigma.2)$ and $(\Sigma.3)$ as in Definition 3.14. By $(\Sigma.2)$ and $(\Sigma.3)$, it is easy to see:

$$K = \{\delta : \exists \mu \ulcorner \delta \mapsto \underline{\mu}^\top \in \Sigma\} = K',$$

and for $\delta \in K$,

$$\begin{aligned} \lambda_\delta &= \text{the unique } \mu \text{ such that } \ulcorner \delta \mapsto \underline{\mu}^\top \in \Sigma = \lambda'_\delta, \\ X_\delta &= \{x : \ulcorner \underline{x} \in \dot{X}_\delta^\top \in \Sigma\} = X'_\delta. \end{aligned}$$

We prove $N = N'$ and $e = e'$ in Claims 3.17 and 3.18, respectively.

Claim 3.17. *For all $\gamma \in \text{Ord}$,*

$$(i) \ V_\gamma^N = V_\gamma^{N'}, \text{ and}$$

$$(ii) \text{ for all } x \in V_\gamma^N, \text{ there is } \alpha_x < \omega_1 \text{ such that } e(\alpha_x) = e'(\alpha_x) = x.$$

Proof of Claim 3.17. We prove this claim by induction. Suppose the claim holds for γ . Fix an $x \in V_{\gamma+1}^N$ and let $\alpha < \omega_1$ be such that $e(\alpha) = x$. We have:

$$\begin{aligned} x &= \{y : y \in x\} \\ &= \{e(\alpha_y) : y \in V_\gamma^N \wedge N \models e(\alpha_y) \in e(\alpha)\} \\ &= \{e(\alpha_y) : y \in V_\gamma^N \wedge \ulcorner \dot{N} \models \dot{\alpha}_y \in \dot{\alpha}^\top \in \Sigma\} \\ &= \{e'(\alpha_y) : y \in V_\gamma^{N'} \wedge \ulcorner \dot{N} \models \dot{\alpha}_y \in \dot{\alpha}^\top \in \Sigma\} \\ &= \{e'(\alpha_y) : y \in V_\gamma^{N'} \wedge N' \models e'(\alpha_y) \in e'(\alpha)\} \\ &\subset e'(\alpha), \end{aligned}$$

where the “=”’s and “ $\subset$ ” are given by $(\Sigma.1)$ and induction hypothesis. On the other hand, if $y \in e'(\alpha)$, let $\beta < \omega_1$ be such that $e'(\beta) = y$. We have:

$$\begin{aligned} N' &\models e'(\beta) \in e'(\alpha) \\ \iff \ulcorner \dot{N} \models \dot{\beta} \in \dot{\alpha}^\top \in \Sigma \\ \iff N &\models e(\beta) \in e(\alpha) = x, \end{aligned}$$

where the “ $\iff$ ”’s are given by $(\Sigma.1)$. Now $e(\beta) \in V_\gamma^N$ and hence by item (ii),

there is an ordinal $\alpha_{e(\beta)} < \omega_1$ such that

$$e(\alpha_{e(\beta)}) = e(\beta).$$

Again by $(\Sigma.1)$,

$$\Vdash \dot{N} \models \dot{\alpha}_{e(\beta)} = \dot{\beta}^\top \in \Sigma,$$

and by item (ii),

$$N' \models e'(\alpha_{e(\beta)}) = e'(\beta).$$

But now

$$y = e'(\beta) = e'(\alpha_{e(\beta)}) = e(\alpha_{e(\beta)}) = e(\beta) \in x,$$

where the third “=” is given by item (ii). This means $x \supset e'(\alpha)$ and hence

$$x = e'(\alpha) \in V_{\gamma+1}^{N'}.$$

We have shown $V_{\gamma+1}^N \subset V_{\gamma+1}^{N'}$. By the same argument $V_{\gamma+1}^N \supset V_{\gamma+1}^{N'}$ also holds. Therefore

$$V_{\gamma+1}^N = V_{\gamma+1}^{N'}.$$

Finally we let $\alpha_x = \alpha$.

□ (Claim 3.17)

Claim 3.18. $e = e'$.

Proof of Claim 3.18. Let $\alpha < \omega_1$. By Claim 3.17,

$$e(\alpha) = e(\alpha_{e(\alpha)}),$$

by $(\Sigma.1)$,

$$\Vdash \dot{N} \models \dot{\alpha} = \dot{\alpha}_{e(\alpha)}^\top \in \Sigma,$$

then by $(\Sigma.1)$ and Claim 3.17 again,

$$e'(\alpha) = e'(\alpha_{e(\alpha)}) = e(\alpha_{e(\alpha)}) = e(\alpha).$$

□ (Claim 3.18, Lemma 3.16)

By Lemma 3.16, if Σ is a syntactic (quasi-)certificate, then we call the unique $\mathfrak{C}$ and e such that Definition 3.14 (or 3.15) is satisfied the corresponding semantic (quasi-)certificate and the corresponding surjection, respectively.

Definition 3.19. Let Σ be an $\mathcal{L}^\lambda$ -syntactic certificate. We say that Σ *certifies* a finite set p of $\mathcal{L}^\lambda$ -formulas if and only if $p \in [\Sigma]^{<\omega}$.

We are now ready to define $\mathbb{P}_\lambda$.

Definition 3.20. $p \in \mathbb{P}_\lambda$ if and only if

$$V^{\text{Col}(\omega, \lambda)} \models \text{There is an } \mathcal{L}^\lambda\text{-syntactic certificate } \Sigma \text{ that certifies } p.$$

The order is defined as $p < q$ if and only if $p \supset q$. We write $\mathbb{P} = \mathbb{P}_\kappa$.

The next absoluteness lemma, which is in essence the same as Lemmas 2.12 and 3.4, is useful and we shall sometimes implicitly make use of it in the rest of the chapter.

Lemma 3.21. *The following are equivalent.*

- (1) $p \in \mathbb{P}_\lambda$,
- (2) for some outer model $W \supset V$ where λ is countable,

$$W \models \text{There is an } \mathcal{L}^\lambda\text{-syntactic certificate } \Sigma \text{ that certifies } p,$$

- (3) for any outer model $W \supset V$ where λ is countable,

$$W \models \text{There is an } \mathcal{L}^\lambda\text{-syntactic certificate } \Sigma \text{ that certifies } p.$$

Proof. The statement “there is an $\mathcal{L}^\lambda$ -syntactic certificate Σ that certifies p ” is Σ_1 in the parameters $\lambda, A, H_{\omega_2}, (A_\mu : \mu \in C \cap \lambda), (Q_\mu : \mu \in C \cap \lambda)$ and $(\mathbb{P}_\mu : \mu \in C \cap \lambda)$ which are all of size at most $|\lambda|$. Then by Shoenfield absoluteness, the statement is absolute across all outer models of V where λ is countable. The lemma follows. $\square$

It is easy to see that the sequence $(\mathbb{P}_\lambda : \lambda \in C \cup \{\kappa\})$ is $\subset$ -increasing; in fact, slightly more can be proved:

Lemma 3.22. *There is a club $D \subset C$ such that*

- (i) $\mathbb{P}_\lambda = \mathbb{P} \cap Q_\lambda$ for all $\lambda \in D$, and
- (ii) $\mathbb{P} = \bigcup_{\lambda \in D} \mathbb{P}_\lambda$.

Proof. Item (ii) is given by (i):

$$\mathbb{P} = \mathbb{P} \cap H_\kappa = \bigcup_{\lambda \in D} (\mathbb{P} \cap Q_\lambda) = \bigcup_{\lambda \in D} \mathbb{P}_\lambda.$$

We prove (i). We first notice that $\mathbb{P}_\lambda \subset Q_\lambda$ for all $\lambda \in C \cup \{\kappa\}$ by our definition and the coding method of the language $\mathcal{L}^\lambda$, cf. Definitions 3.12 and 3.13. The rest of the proof consists of the following claims.

Claim 3.23. *The sequence $(\mathbb{P}_\lambda : \lambda \in C \cup \{\kappa\})$ is $\subset$ -increasing, i.e. $\mathbb{P}_\mu \subset \mathbb{P}_\lambda$ for all $\mu \leq \lambda$ in $C \cup \{\kappa\}$. In particular, $\mathbb{P}_\lambda \subset \mathbb{P}$ for all $\lambda \in C$.*

Proof of Claim 3.23. Fix $\mu \leq \lambda$ in $C \cup \{\kappa\}$ and let $p \in \mathbb{P}_\mu$. Then

$$V^{\text{Col}(\omega, \mu)} \models \text{There is an } \mathcal{L}^\mu\text{-syntactic certificate } \Sigma \text{ that certifies } p,$$

and by Lemma 3.21,

$$V^{\text{Col}(\omega, \lambda)} \models \text{There is an } \mathcal{L}^\mu\text{-syntactic certificate } \Sigma \text{ that certifies } p.$$

Fix an $\mathcal{L}^\mu$ -syntactic certificate $\Sigma \in V^{\text{Col}(\omega, \lambda)}$ that certifies p , and let

$$\mathfrak{C} = (N, (\lambda_\delta, X_\delta : \delta \in K))$$

and e , respectively, be the corresponding μ -semantic certificate and surjection. Σ is also a set of $\mathcal{L}^\lambda$ -formulas as $\mathcal{L}^\mu \subset \mathcal{L}^\lambda$. By checking through conditions $(\Sigma.1)$ – $(\Sigma.4)$ as in Definition 3.14 and $(\mathfrak{C}.1)$ – $(\mathfrak{C}.5)$ as in Definition 3.10, we have that Σ is an $\mathcal{L}^\lambda$ -syntactic certificate and that $\mathfrak{C}$ is a λ -semantic certificate. This means

$$V^{\text{Col}(\omega, \lambda)} \models \text{There is an } \mathcal{L}^\lambda\text{-syntactic certificate } \Sigma \text{ that certifies } p,$$

hence $p \in \mathbb{P}_\lambda$. □ (Claim 3.23)

By Claim 3.23, $\mathbb{P}_\lambda \subset \mathbb{P} \cap Q_\lambda$ for all $\lambda \in C$. The other inclusion is a bit more subtle.

Claim 3.24. $\mathbb{P} \cap Q_\lambda \subset \mathbb{P}_\lambda$ for unboundedly many $\lambda \in C$.

Proof of Claim 3.24. We inductively define a sequence $(\lambda_n : n < \omega)$ of ordinals below κ such that $\lambda_n \in C$ for each $n < \omega$. Let λ_0 be any ordinal in C . Suppose

that λ_n has already been defined, we aim to define λ_{n+1} . For $p \in \mathbb{P} \cap Q_{\lambda_n}$, we pick an ordinal λ^p such that

$$\begin{aligned} \text{Col}(\omega, \kappa) \Vdash & \text{“for some } \mathcal{L}^\kappa\text{-syntactic certificate } \Sigma \text{ and its corresponding} \\ & \kappa\text{-semantic certificate } \mathfrak{C} = (N, (\lambda'_\delta, X'_\delta : \delta \in K)), \\ & \Sigma \text{ certifies } p \text{ and } \text{Ord}^N = \lambda^p\text{.”} \end{aligned}$$

By $(\mathfrak{C}.2)$, we have $\lambda^p < \kappa$. We note that the definition of this λ^p is carried out in V as the forcing language is definable. Recall that $\kappa = (2^{\aleph_1})^+V$ is a regular uncountable cardinal in V . Then we let λ_{n+1} be any ordinal in C such that

$$\lambda_{n+1} > \max\{\lambda_n, \sup\{\lambda^p : p \in \mathbb{P} \cap Q_{\lambda_n}\}\}.$$

Finally, we let

$$\lambda = \sup\{\lambda_n : n < \omega\}.$$

Then $\lambda < \kappa$ as the sequence $(\lambda_n : n < \omega)$ is produced in V and the cofinality of κ is uncountable in V ; and $\lambda \in C$ as C is a club and $\lambda_n \in C$ for each $n < \omega$.

We remain to show that $\mathbb{P} \cap Q_\lambda \subset \mathbb{P}_\lambda$. Fix $p \in \mathbb{P} \cap Q_\lambda$ and let $\Sigma \in V^{\text{Col}(\omega, \kappa)}$ be an $\mathcal{L}^\kappa$ -syntactic certificate that certifies p , and let

$$\mathfrak{C} = (N, (\lambda'_\delta, X'_\delta : \delta \in K))$$

and e be the corresponding κ -semantic certificate and surjection respectively. We may assume $\text{Ord}^N = \lambda^p$. Then $\lambda^p < \lambda$. We let

$$\begin{aligned} \Sigma' &= \Sigma \cap Q_\lambda, \\ K' &= \{\delta \in K : \lambda'_\delta < \lambda\}, \text{ and} \\ \mathfrak{C}' &= (N, (\lambda'_\delta, X'_\delta : \delta \in K')). \end{aligned}$$

Again, checking through conditions $(\mathfrak{C}.1)$ – $(\mathfrak{C}.5)$ as in Definition 3.10 and $(\Sigma.1)$ – $(\Sigma.4)$ as in Definition 3.14 gives us that Σ' is an $\mathcal{L}^\lambda$ -syntactic certificate with $\mathfrak{C}'$ and e being respectively the corresponding λ -semantic certificate and surjection. (As for $(\mathfrak{C}.2)$, we simply notice that $\text{Ord}^N = \lambda^p < \lambda$.) This means

$$V^{\text{Col}(\omega, \kappa)} \models \Sigma' \text{ is an } \mathcal{L}^\lambda\text{-syntactic certificate and } \Sigma' \text{ certifies } p.$$

By Lemma 3.21, $p \in \mathbb{P}_\lambda$.

□ (Claim 3.24)

Claim 3.24 together with the next claim shows us that the set of all λ 's such that $\mathbb{P} \cap Q_\lambda \subset \mathbb{P}_\lambda$ contains a club. We finish the proof of the current lemma by proving:

Claim 3.25. *Let $\gamma < \kappa$ be a limit ordinal and let $(\lambda_\xi : \xi < \gamma)$ be an increasing sequence of ordinals in V such that $\lambda_\xi \in C$ and $\mathbb{P} \cap Q_{\lambda_\xi} \subset \mathbb{P}_{\lambda_\xi}$ for all $\xi < \gamma$. Then $\lambda \in C$ and $\mathbb{P} \cap Q_\lambda \subset \mathbb{P}_\lambda$, where $\lambda = \bigcup_{\xi < \gamma} \lambda_\xi$.*

Proof of Claim 3.25. $\lambda \in C$ since C is a club. Then

$$\mathbb{P} \cap Q_\lambda = \bigcup_{\xi < \gamma} (\mathbb{P} \cap Q_{\lambda_\xi}) \subset \bigcup_{\xi < \gamma} \mathbb{P}_{\lambda_\xi} \subset \mathbb{P}_\lambda,$$

where the first “ $\subset$ ” is given by the assumption of this claim, and the second “ $\subset$ ” is given by Claim 3.23. □ (Claim 3.25, Lemma 3.22)

Just the same as [3, Lemma 3.6], we are happy if our recursively defined forcings are not empty.

Lemma 3.26. $\emptyset \in \mathbb{P}_{\min(C)}$.

Proof. Write $\lambda = \min(C)$ and work in $V^{\text{Col}(\omega, \lambda)}$. Recall that $\varphi(A)$ is good and weakly honestly consistent. By Lemma 3.4, let N be a countable witness for the weak honest consistency of “ $\varphi(A) \wedge \text{cf } \omega_2^V = \omega$ ” such that $H_{\omega_2}^V$ has size $\aleph_1$ in N and the ordinal height of N is η_A which is less than λ by (Q.1) as in Lemma 3.6. Let

$$\mathfrak{C} = (N, ())$$

and let $e : \omega_1 \rightarrow N$ be a surjection. Thus conditions $(\mathfrak{C}.1)$ and $(\mathfrak{C}.2)$ hold. Conditions $(\mathfrak{C}.3)$ – $(\mathfrak{C}.5)$ are vacuously true as the second component of $\mathfrak{C}$ is empty.

It is straightforward to derive a set of $\mathcal{L}^\lambda$ -formulas Σ from $\mathfrak{C}$ and e such that condition $(\Sigma.1)$ is satisfied by $\Sigma, \mathfrak{C}$ and e . Again, conditions $(\Sigma.2)$ – $(\Sigma.4)$ are vacuously true. This means that Σ is a syntactic certificate and it certifies $\emptyset$. Hence $\emptyset \in \mathbb{P}_\lambda$. □

3.3 Deriving a quasi-certificate from a filter

The only task in this section is to prove the following lemma.

Lemma 3.27. *Let $\lambda \in C \cup \{\kappa\}$. Let g be a filter for $\mathbb{P}_\lambda$ such that $g \cap E \neq \emptyset$ for all dense $E \subset \mathbb{P}_\lambda$ which are definable over $(Q_\lambda; \in, \mathbb{P}_\lambda)$ using parameters in Q_λ . Then $\bigcup g$ is a syntactic quasi-certificate (cf. Definition 3.15).*

Proof. Define

$$\text{Terms} = \{\dot{\alpha} : \alpha < \omega_1\} \cup \{\underline{x} : x \in H_{\omega_2}^V\} \cup \{\underline{A}, \underline{\omega_2}^V, \underline{H_{\omega_2}^V}\}.$$

For $\tau, \sigma \in \text{Terms}$, we define

$$\begin{aligned} \tau \sim \sigma & \quad \text{iff } \ulcorner \dot{N} \models \tau = \sigma \urcorner \in \bigcup g \\ [\tau] & = \{\sigma : \tau \sim \sigma\} \\ N & = \{[\tau] : \tau \in \text{Terms}\} \\ [\tau] \tilde{\in} [\sigma] & \quad \text{iff } \ulcorner \dot{N} \models \tau \in \sigma \urcorner \in \bigcup g. \end{aligned}$$

We will show that N is a witness for the weak honest consistency of $\varphi(A)$. Recall that N also has a predicate for the nonstationary ideal, cf. Definition 2.11; naturally we let

$$\widetilde{\text{NS}}_{\omega_1} = \{[\tau] : \tau \in \text{Terms} \text{ and } \ulcorner \dot{N} \models \tau \in \text{NS}_{\omega_1} \urcorner \in \bigcup g\}^{14}$$

We also let

$$K = \{\delta : \exists \mu \ulcorner \dot{\delta} \mapsto \underline{\mu} \urcorner \in \bigcup g\}^{15}$$

and for $\delta \in K$,

$$\begin{aligned} \lambda_\delta & = \text{the unique } \mu \text{ such that } \ulcorner \dot{\delta} \mapsto \underline{\mu} \urcorner \in \bigcup g, \text{ and} \\ X_\delta & = \{x : \ulcorner \underline{x} \in \dot{X}_\delta \urcorner \in \bigcup g\}. \end{aligned}$$

¹⁴ $\widetilde{\text{NS}}_{\omega_1}$ is understood as the predicate for the nonstationary ideal, cf. Definition 3.12.

¹⁵If $\delta \in K$, then there is a unique μ such that $\ulcorner \dot{\delta} \mapsto \underline{\mu} \urcorner \in \bigcup g$; the uniqueness of such μ will be verified in Lemma 3.28.

Finally, we let

$$\mathfrak{C} = (N, (\lambda_\delta, X_\delta : \delta \in K)),^{16}$$

and for $\alpha < \omega_1$,

$$e(\alpha) = [\dot{\alpha}].$$

We will prove this lemma by running (quite) a few density arguments. However, the whole proof is rather lengthy and we shall break it down into several smaller lemmas:

- $\mathfrak{C}$ satisfies condition $(\mathfrak{C}.1)$ as in Definition 3.10 by Lemma 3.29,
- $\mathfrak{C}$ satisfies $(\mathfrak{C}.3)$ simply because it makes no sense by the definition of $\mathcal{L}^\lambda$ if it is not the case $\delta < \omega_1$, cf. Definition 3.12,
- $\mathfrak{C}$ satisfies $(\mathfrak{C}.4)$ and $(\mathfrak{C}.5)$ by Lemma 3.37, hence $\mathfrak{C}$ is a semantic quasi-certificate,
- e is a surjection by Lemma 3.40,
- $\bigcup g, \mathfrak{C}$ and e satisfy condition $(\Sigma.1)$ as in Definition 3.14 by Claim 3.32,
- $\bigcup g$ and $\mathfrak{C}$ satisfy $(\Sigma.2)$ and $(\Sigma.3)$ by our construction of K and λ_δ, X_δ for $\delta \in K$,
- $\bigcup g$ and $\mathfrak{C}$ satisfy $(\Sigma.4)$ by Lemma 3.41, hence $\bigcup g$ is indeed a syntactic quasi-certificate. $\square$

We start proving the lemmas and claims. As a warm-up, we first show that λ_δ is well-defined for $\delta \in K$:

Lemma 3.28. *Let $\delta \in K$. Then there is a unique μ such that*

$$\ulcorner \underline{\delta} \mapsto \underline{\mu} \urcorner \in \bigcup g.$$

Proof of Lemma 3.28. The existence of μ is given by the definition of K . We prove the uniqueness of μ . Suppose that μ_1, μ_2 are such that

$$\{\ulcorner \underline{\delta} \mapsto \underline{\mu}_1 \urcorner, \ulcorner \underline{\delta} \mapsto \underline{\mu}_2 \urcorner\} \subset \bigcup g.$$

¹⁶Here we identify N with the transitive collapse of $(N; \tilde{\in}, \tilde{\text{NS}}_{\omega_1})$. The well-foundedness of $(N; \tilde{\in}, \tilde{\text{NS}}_{\omega_1})$ will be proved later.

Fix a $p \in g$ such that

$$\{\ulcorner \underline{\delta} \mapsto \underline{\mu_1} \urcorner, \ulcorner \underline{\delta} \mapsto \underline{\mu_2} \urcorner\} \subset p.$$

Let Σ' certify p and let

$$\mathfrak{C}' = (N', (\lambda'_\xi, X'_\xi : \xi \in K'))$$

be the corresponding semantic certificate. As Σ' certifies p , we have

$$\{\ulcorner \underline{\delta} \mapsto \underline{\mu_1} \urcorner, \ulcorner \underline{\delta} \mapsto \underline{\mu_2} \urcorner\} \subset \Sigma'.$$

Then by $(\Sigma.2)$ we have $\delta \in K'$ and

$$\mu_1 = \mu_2 = \lambda'_\delta. \quad \square \text{ (Lemma 3.28)}$$

Lemma 3.29. $\mathfrak{C}$ satisfies condition $(\mathfrak{C}.1)$ as in Definition 3.10.

Proof. The proof of Lemma 3.29 consists of Claims 3.30 – 3.36. Let us temporarily define the interpretation of elements of Terms for this lemma. Suppose that Σ' is a syntactic certificate, and

$$\mathfrak{C}' = (N', (\lambda'_\delta, X'_\delta : \delta \in K'))$$

and $e' : \omega_1 \rightarrow N'$ are the corresponding semantic certificate and surjection, respectively, then for $\tau \in \text{Terms}$, the Σ' -interpretation of τ , denoted by $\tau^{\Sigma'}$, is defined as

$$\begin{aligned} \tau^{\Sigma'} &= e'(\alpha), \text{ if } \tau = \dot{\alpha} \text{ for some } \alpha < \omega_1, \text{ or} \\ \tau^{\Sigma'} &= x, \text{ if } \tau = \underline{x} \text{ for some } x \in H_{\omega_2}^V \cup \{A, \omega_2^V, H_{\omega_2}^V\}. \end{aligned}$$

This notation will be used in some claims. Let us now start proving the lemma. As a warm-up, we first verify that the “ $\widetilde{\in}$ ” relation and the “ $\widetilde{\text{NS}}_{\omega_1}$ ” predicate are well-defined.

Claim 3.30. *Let $\tau_1, \tau_2, \sigma_1, \sigma_2 \in \text{Terms}$ be such that $\tau_1 \sim \tau_2$ and $\sigma_1 \sim \sigma_2$. Then*

$$\begin{aligned} \ulcorner \dot{N} \models \tau_1 \in \sigma_1 \urcorner \in \bigcup g &\iff \ulcorner \dot{N} \models \tau_2 \in \sigma_2 \urcorner \in \bigcup g, \text{ and} \\ \ulcorner \dot{N} \models \tau_1 \in \dot{N}S_{\omega_1} \urcorner \in \bigcup g &\iff \ulcorner \dot{N} \models \tau_2 \in \dot{N}S_{\omega_1} \urcorner \in \bigcup g. \end{aligned}$$

Proof of Claim 3.30. We only give a proof for the forward direction of the first “ $\iff$ ” as the other direction and the second “ $\iff$ ” can be shown by the same argument. By assumption we have

$$\{\ulcorner \dot{N} \models \tau_1 = \tau_2 \urcorner, \ulcorner \dot{N} \models \sigma_1 = \sigma_2 \urcorner\} \subset \bigcup g.$$

Suppose

$$\ulcorner \dot{N} \models \tau_1 \in \sigma_1 \urcorner \in \bigcup g$$

and let $p \in g$ be such that

$$\{\ulcorner \dot{N} \models \tau_1 = \tau_2 \urcorner, \ulcorner \dot{N} \models \sigma_1 = \sigma_2 \urcorner, \ulcorner \dot{N} \models \tau_1 \in \sigma_1 \urcorner\} \subset p,$$

and let $p' < p$. Let Σ' certify p' and let

$$\mathfrak{C}' = (N', (\lambda'_\delta, X'_\delta : \delta \in K'))$$

be the corresponding semantic certificate. By $(\Sigma.1)$,

$$N' \models \tau_1^{\Sigma'} = \tau_2^{\Sigma'} \wedge \sigma_1^{\Sigma'} = \sigma_2^{\Sigma'} \wedge \tau_1^{\Sigma'} \in \sigma_1^{\Sigma'},$$

and hence

$$N' \models \tau_2^{\Sigma'} \in \sigma_2^{\Sigma'}.$$

By $(\Sigma.1)$ again,

$$\ulcorner \dot{N} \models \tau_2 \in \sigma_2 \urcorner \in \Sigma',$$

which means

$$p' \cup \{\ulcorner \dot{N} \models \tau_2 \in \sigma_2 \urcorner\}$$

is a condition in $\mathbb{P}_\lambda$ stronger than p' as it is also certified by Σ' . We have shown that the set

$$D_{\bar{\epsilon}} = \{q \in \mathbb{P}_\lambda : \ulcorner \dot{N} \models \tau_2 \in \sigma_2 \urcorner \in q\}$$

is dense below p . Clearly, $D_{\tilde{\epsilon}}$ is definable over $(Q_\lambda; \in, \mathbb{P}_\lambda)$ using parameters from Q_λ . Let $q \in g \cap D_{\tilde{\epsilon}}$. Then q witnesses

$$\ulcorner \dot{N} \models \tau_2 \in \sigma_2 \urcorner \in \bigcup g. \quad \square \text{ (Claim 3.30)}$$

The “ $\sim$ ” being an equivalence relation can be shown in exactly the same way and we shall omit the proof. Let us turn to proving properties of N .

Claim 3.31. *Let $x \in H_{\omega_2}^V \cup \{A, \omega_2^V, H_{\omega_2}^V\}$. Then $[x] = x$.¹⁷*

Proof of Claim 3.31. First let $y \in x$. Then

$$D_{y \in x} = \{q \in \mathbb{P}_\lambda : \ulcorner \dot{N} \models \underline{y} \in \underline{x} \urcorner \in q\}$$

is a dense subset of $\mathbb{P}_\lambda$ simply because $\{\ulcorner \dot{N} \models \underline{y} \in \underline{x} \urcorner\}$ is certified by every syntactic certificate, hence

$$\ulcorner \dot{N} \models \underline{y} \in \underline{x} \urcorner \in \bigcup g$$

and $[y] \tilde{\epsilon} [x]$ by definition.

On the other hand, let $\tau \in \text{Terms}$ be such that $[\tau] \tilde{\epsilon} [x]$, we show that there is $y \in x$ such that $[\tau] = [y]$. Let $p \in g$ be such that

$$\ulcorner \dot{N} \models \tau \in \underline{x} \urcorner \in p$$

and let $p' < p$. Let Σ' certify p' and let

$$\mathfrak{C}' = (N', (\lambda'_\xi, X'_\xi : \xi \in K'))$$

be the corresponding semantic certificate. Then

$$N' \models \tau^{\Sigma'} \in x,$$

and

$$p' \cup \{\ulcorner \dot{N} \models \tau = \underline{\tau^{\Sigma'}} \urcorner\}$$

¹⁷In the statement of this claim we have already implicitly assumed the well-foundedness of the structure $(\{[x] : x \in H_{\omega_2}^V\} \cup \{[A], [\omega_2^V], [H_{\omega_2}^V]\}; \tilde{\epsilon})$ and hence it is identified with its transitive collapse. The well-foundedness of this structure will be clear in the proof.

is certified by Σ' , and hence the set

$$D_{\tau=y} = \{q \in \mathbb{P}_\lambda : \exists y \in x \text{ } \dot{\Vdash} \tau = \underline{y}^\top \in q\}$$

is dense below p . This means $[\tau] = [\underline{y}]$ for some $y \in x$.

By the above arguments, for any $x, y \in H_{\omega_2}^V \cup \{A, \omega_2^V, H_{\omega_2}^V\}$,

$$y \in x \iff [\underline{y}] \tilde{\in} [\underline{x}].$$

This means that the “ $\tilde{\in}$ ” relation is well-founded on

$$\{[\underline{x}] : x \in H_{\omega_2}^V\} \cup \{[\underline{A}], [\underline{\omega_2^V}], [\underline{H_{\omega_2}^V}]\}.$$

Now we show that $[\emptyset]$ is the empty set of $(N; \tilde{\in})$. It is easy to see that $\{\dot{\Vdash} \tau \in \underline{\emptyset}^\top\}$ can never be certified, which means

$$\dot{\Vdash} \tau \in \underline{\emptyset}^\top \notin \bigcup g$$

for any $\tau \in \text{Terms}$.

The claim follows by running a typical induction on the rank of x . $\square$ (Claim 3.31)

Claim 3.32. *Let Φ be a formula in $\mathcal{L}_{\dot{\in}, \widetilde{\text{NS}}_{\omega_1}}$ and let $\tau_1, \dots, \tau_k \in \text{Terms}$. Then*

$$(N; \tilde{\in}, \widetilde{\text{NS}}_{\omega_1}) \models \Phi([\tau_1], \dots, [\tau_k]) \iff \dot{\Vdash} \Phi(\tau_1, \dots, \tau_k)^\top \in \bigcup g.$$

Proof of Claim 3.32. We prove this claim by induction on the complexity of Φ . The proof of Claim 3.30 actually shows that this claim holds for atomic Φ . For $\Phi \equiv \Phi_0 \wedge \Phi_1$ and $\Phi \equiv \neg \Psi$ this claim can be shown by easy analogues of the proof for $\Phi \equiv \exists v \Psi(v)$. We only give a proof for the case when $\Phi \equiv \exists v \Psi(v)$.

We prove:

$$\begin{aligned} & (N; \tilde{\in}, \widetilde{\text{NS}}_{\omega_1}) \models \exists v \Psi(v, [\tau_1], \dots, [\tau_k]) \\ \iff & \exists \sigma \in \text{Terms}, (N; \tilde{\in}, \widetilde{\text{NS}}_{\omega_1}) \models \Psi([\sigma], [\tau_1], \dots, [\tau_k]) \\ \iff & \exists \sigma \in \text{Terms}, \dot{\Vdash} \Psi(\sigma, \tau_1, \dots, \tau_k)^\top \in \bigcup g \\ \iff & \dot{\Vdash} \exists v \Psi(v, \tau_1, \dots, \tau_k)^\top \in \bigcup g. \end{aligned}$$

The first “ $\Longleftrightarrow$ ” is completely trivial. The second “ $\Longleftrightarrow$ ” is given by induction hypothesis. Now we prove the forward direction of the third “ $\Longleftrightarrow$ ”. Fix $\sigma \in \text{Terms}$ and let $p \in g$ be such that

$$\ulcorner \dot{N} \models \Psi(\sigma, \tau_1, \dots, \tau_k)^\top \in p,$$

and let $p' < p$. Let Σ' certify p' and let

$$\mathfrak{C}' = (N', (\lambda'_\delta, X'_\delta : \delta \in K'))$$

be the corresponding semantic certificate. Then by $(\Sigma.1)$,

$$N' \models \Psi(\sigma^{\Sigma'}, \tau_1^{\Sigma'}, \dots, \tau_k^{\Sigma'}),$$

and trivially

$$N' \models \exists v \Psi(v, \tau_1^{\Sigma'}, \dots, \tau_k^{\Sigma'}).$$

By $(\Sigma.1)$ again, this means that

$$p' \cup \{\ulcorner \dot{N} \models \exists v \Psi(v, \tau_1, \dots, \tau_k)^\top\}$$

is certified by Σ' . Therefore the set

$$D_{\tau_1, \dots, \tau_k}^1 = \{q \in \mathbb{P}_\lambda : \ulcorner \dot{N} \models \exists v \Psi(v, \tau_1, \dots, \tau_k)^\top \in q\}$$

is dense below p and hence

$$\ulcorner \dot{N} \models \exists v \Psi(v, \tau_1, \dots, \tau_k)^\top \in \bigcup g.$$

As for the backward direction, let $p \in g$ be such that

$$\ulcorner \dot{N} \models \exists v \Psi(v, \tau_1, \dots, \tau_k)^\top \in p,$$

and let $p' < p$. Let Σ' certify p' and let

$$\mathfrak{C}' = (N', (\lambda'_\delta, X'_\delta : \delta \in K'))$$

and $e' : \omega_1 \rightarrow N'$ be the corresponding semantic certificate and surjection respec-

tively. Then

$$N' \models \exists v \Psi(v, \tau_1^{\Sigma'}, \dots, \tau_k^{\Sigma'}),$$

and since e' is surjective, for some $\alpha < \omega_1$,

$$N' \models \Psi(e'(\alpha), \tau_1^{\Sigma'}, \dots, \tau_k^{\Sigma'}).$$

Then

$$p' \cup \{\ulcorner \dot{N} \models \Psi(\dot{\alpha}, \tau_1, \dots, \tau_k) \urcorner\}$$

is certified by Σ' , and the set

$$D_{\tau_1, \dots, \tau_k}^2 = \{q \in \mathbb{P}_\lambda : \exists \alpha < \omega_1 \ulcorner \dot{N} \models \Psi(\dot{\alpha}, \tau_1, \dots, \tau_k) \urcorner \in q\}$$

is dense below p and hence for some $\sigma \in \text{Terms}$,

$$\ulcorner \dot{N} \models \Psi(\sigma, \tau_1, \dots, \tau_k) \urcorner \in \bigcup g. \quad \square \text{ (Claim 3.32)}$$

We point out that Claim 3.32 also verifies $(\Sigma.1)$ once the necessary properties are proved for $\mathfrak{C}$ and e .

Claim 3.33. *Let Φ be a formula in $\mathcal{L}_{\dot{\epsilon}, \text{NS}_{\omega_1}}$ and let $x_1, \dots, x_k \in H_{\omega_2}^V \cup \{A, \omega_2^V, H_{\omega_2}^V\}$. Suppose that*

$$N' \models \Phi(x_1, \dots, x_k)$$

for every semantic certificate

$$\mathfrak{C}' = (N', (\lambda'_\delta, X'_\delta : \delta \in K')).$$

Then

$$\ulcorner \dot{N} \models \Phi(\underline{x}_1, \dots, \underline{x}_k) \urcorner \in \bigcup g.$$

Proof of Claim 3.33. Fix $p \in \mathbb{P}_\lambda$. Let Σ' certify p and let

$$\mathfrak{C}' = (N', (\lambda'_\delta, X'_\delta : \delta \in K'))$$

be the corresponding semantic certificate. Then

$$N' \models \Phi(x_1, \dots, x_k),$$

by our assumption and hence

$$p \cup \{\ulcorner \dot{N} \models \Phi(\underline{x}_1, \dots, \underline{x}_k) \urcorner\}$$

is also certified by Σ' . We have shown that the set

$$D_{x_1, \dots, x_k} = \{q \in \mathbb{P}_\lambda : \ulcorner \dot{N} \models \Phi(\underline{x}_1, \dots, \underline{x}_k) \urcorner \in q\}$$

is dense in $\mathbb{P}_\lambda$. Then

$$\ulcorner \dot{N} \models \Phi(\underline{x}_1, \dots, \underline{x}_k) \urcorner \in \bigcup g. \quad \square \text{ (Claim 3.33)}$$

Claim 3.34. *Let Φ be a formula in $\mathcal{L}_{\dot{N}, \widetilde{\text{NS}}_{\omega_1}}$ and let $\tau_1, \dots, \tau_k \in \text{Terms}$. Suppose that*

$$\text{ZFC}^- \vdash \Phi_0(v_1, \dots, v_k) \iff \Phi_1(v_1, \dots, v_k).$$

Then

$$\ulcorner \dot{N} \models \Phi_0(\tau_1, \dots, \tau_k) \urcorner \in \bigcup g \iff \ulcorner \dot{N} \models \Phi_1(\tau_1, \dots, \tau_k) \urcorner \in \bigcup g.$$

Proof of Claim 3.34. This claim can be shown by the same proof of Claim 3.33. We shall omit the details. $\square$ (Claim 3.34)

With the help of the above claims, we can show that $\widetilde{\text{NS}}_{\omega_1}$ is indeed the nonstationary ideal of N .

Claim 3.35. $\widetilde{\text{NS}}_{\omega_1} = \{[\tau] : (N; \widetilde{\epsilon}, \widetilde{\text{NS}}_{\omega_1}) \models [\tau] \text{ is nonstationary}\}.$

Proof of Claim 3.35. Let $\tau \in \text{Terms}$. Then

$$\begin{aligned} [\tau] \in \widetilde{\text{NS}}_{\omega_1} &\iff \ulcorner \dot{N} \models \tau \in \dot{\text{NS}}_{\omega_1} \urcorner \in \bigcup g \\ &\iff \ulcorner \dot{N} \models \tau \text{ is nonstationary} \urcorner \in \bigcup g \\ &\iff (N; \widetilde{\epsilon}, \widetilde{\text{NS}}_{\omega_1}) \models [\tau] \text{ is nonstationary,} \end{aligned}$$

where the first “ $\iff$ ” is by definition, the second “ $\iff$ ” is given by Claim 3.34, and the third “ $\iff$ ” is given by Claim 3.32, $\square$ (Claim 3.35)

Claims 3.31, 3.32, 3.33, and 3.35 together verify that $(N; \widetilde{\epsilon}, \widetilde{\text{NS}}_{\omega_1})$ witnesses the weak honest consistency of “ $\varphi(A) \wedge \text{cf } \omega_2^V = \omega$ ” except for that N is yet to be

shown transitive, and that $H_{\omega_2}^V$ has size $\aleph_1$ in N . In order to prove that $(\mathfrak{C}.1)$ is satisfied by N , we remain to show the well-foundedness of $(N; \widetilde{\in}, \widetilde{\text{NS}}_{\omega_1})$, which is given by the following claim.

Claim 3.36. *Suppose $(N; \widetilde{\in}, \widetilde{\text{NS}}_{\omega_1}) \models [\tau] \in \text{Ord}$. Then $[\tau] = \xi$ for some $\xi < \lambda$. In particular, $(N; \widetilde{\in}, \widetilde{\text{NS}}_{\omega_1})$ is well-founded.*

Proof of Claim 3.36. By Claim 3.32, let $p \in g$ be such that

$$\ulcorner \dot{N} \models \tau \in \text{Ord} \urcorner \in p$$

and let $p' < p$. Let Σ' certify p' and let $\mathfrak{C}' = (N', (\lambda'_\delta, X'_\delta : \delta \in K'))$ be the corresponding semantic certificate. As N' has ordinal height less than λ , there is some $\xi < \lambda$ such that

$$N' \models \tau^{\Sigma'} = \xi.$$

This means that

$$p' \cup \{\ulcorner \dot{N} \models \tau = \underline{\xi} \urcorner\}$$

is also certified by Σ' . We have shown that

$$D_{\tau=\xi} = \{q \in \mathbb{P}_\lambda : \exists \xi < \lambda \ulcorner \dot{N} \models \tau = \underline{\xi} \urcorner \in q\}$$

is dense below p and definable over $(Q_\lambda; \in, \mathbb{P}_\lambda)$ using parameters in Q_λ . Hence

$$\ulcorner \dot{N} \models \tau = \underline{\xi} \urcorner \in \bigcup g$$

for some $\xi < \lambda$, and by Claim 3.32 again, $[\tau] = \xi$. $\square$ (Claim 3.36, Lemma 3.29)

Lemma 3.37. $\mathfrak{C}$ satisfies conditions $(\mathfrak{C}.4)$ and $(\mathfrak{C}.5)$ as in Definition 3.10.

Proof. Fix $\delta \in K$. Then by definition λ_δ is the unique ordinal such that

$$\ulcorner \underline{\delta} \mapsto \underline{\lambda_\delta} \urcorner \in \bigcup g.$$

Claim 3.38. $\lambda_\delta \in C \cap \lambda$.

Proof of Claim 3.38. Let $p \in g$ be such that

$$\ulcorner \underline{\delta} \mapsto \underline{\lambda_\delta} \urcorner \in p.$$

Let Σ' certify p and let $\mathfrak{C}' = (N', (\lambda'_\eta, X'_\eta : \eta \in K'))$ be the corresponding semantic certificate. Since

$$\ulcorner \underline{\delta} \mapsto \underline{\lambda_\delta} \urcorner \in \Sigma',$$

by $(\Sigma.2)$ we have $\delta \in K'$ and $\lambda_\delta = \lambda'_\delta$; then by $(\mathfrak{C}.4)$, $\lambda'_\delta \in C \cap \lambda$. $\square$ (Claim 3.38)

As for the second part of $(\mathfrak{C}.4)$, we shall omit the proof since it is similar to Claim 3.38. We verify $(\mathfrak{C}.5)$ and thus complete the proof of Lemma 3.37. Again, we shall only verify the first part of $(\mathfrak{C}.5)$.

Claim 3.39. $X_\delta \prec (Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta})$.

Proof of Claim 3.39. Let Φ be a formula and let $x_1, \dots, x_k \in X_\delta$ be such that

$$(Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta}) \models \exists v \Phi(v, x_1, \dots, x_k).^{18}$$

Let $p \in g$ be such that

$$\{\ulcorner \underline{\delta} \mapsto \underline{\lambda_\delta} \urcorner, \ulcorner \underline{x_1} \in \dot{X}_\delta \urcorner, \dots, \ulcorner \underline{x_k} \in \dot{X}_\delta \urcorner\} \subset p$$

and let $p' < p$. Let Σ' certify p' and let $\mathfrak{C}' = (N', (\lambda'_\eta, X'_\eta : \eta \in K'))$ be the corresponding semantic certificate. Then by $(\mathfrak{C}.5)$ and $(\Sigma.3)$,

$$\{x_1, \dots, x_k\} \subset X'_\delta \prec (Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta}).$$

Clearly there is some $y \in X'_\delta$ be such that

$$(Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta}) \models \Phi(y, x_1, \dots, x_k),$$

and for this y ,

$$p' \cup \{\ulcorner \underline{y} \in \dot{X}_\delta \urcorner\} \in \mathbb{P}_\lambda$$

as being certified by Σ' . We have shown that the following set

$$D_{\prec} = \{q \in \mathbb{P}_\lambda : \exists y \in Q_{\lambda_\delta} (\ulcorner \underline{y} \in \dot{X}_\delta \urcorner \in q \wedge (Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta}) \models \Phi(y, x_1, \dots, x_k))\}$$

¹⁸Here we omit the proof that $X_\delta \subset Q_{\lambda_\delta}$ because it is also very similar to Claim 3.38.

is dense below p . Then there is some $y \in X_\delta$ such that

$$(Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta}) \models \Phi(y, x_1, \dots, x_k),$$

and by Tarski–Vaught test,

$$X_\delta \prec (Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta}).$$

□ (Claim 3.39, Lemma 3.37)

Lemma 3.40. *For every $x \in H_{\omega_2}^V \cup \{\omega_2^V, H_{\omega_2}^V, A\}$, there is $\alpha < \omega_1$ such that $\dot{\alpha} \sim \underline{x}$, i.e. $\dot{N} \models \dot{\alpha} = \underline{x}^\top \in \bigcup g$. In particular, $e : \omega_1 \rightarrow N$ is a surjection.*

Proof. Let $x \in H_{\omega_2}^V \cup \{\omega_2^V, H_{\omega_2}^V, A\}$. The following set

$$D_x = \{q \in \mathbb{P}_\lambda : \dot{N} \models \underline{x} = \underline{x}^\top \in q\}$$

is obviously dense. Let $p \in D_x \cap g$, let Σ' certifies p , and let

$$\mathfrak{C}' = (N', (\lambda'_\delta, X'_\delta : \delta \in K'))$$

and $e' : \omega_1 \rightarrow N'$ be respectively the corresponding semantic certificate and surjection. Since e' is surjective, we have that $e'(\alpha) = x$ for some $\alpha < \omega_1$. This means that

$$D_{\alpha, x} = \{q \in \mathbb{P}_\lambda : \dot{N} \models \dot{\alpha} = \underline{x}^\top \in q\}$$

is dense below p . The lemma follows. □

We remain to verify $(\Sigma.4)$.

Lemma 3.41. $\bigcup g$ and $\mathfrak{C}$ satisfy condition $(\Sigma.4)$ as in Definition 3.14.

Proof. Fix $\delta \in K$. Let $x_1, \dots, x_k \in X_\delta$ be such that if we write

$$E = \{r \in \mathbb{P}_{\lambda_\delta} : (Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta}) \models \Phi(r, x_1, \dots, x_k)\},$$

then E is dense in $\mathbb{P}_{\lambda_\delta}$. Let $p \in g$ be such that

$$\{\dot{\lambda}_\delta \mapsto \underline{\lambda}_\delta^\top, \dot{x}_1 \mapsto \underline{x}_1^\top \in \dot{X}_\delta^\top, \dots, \dot{x}_k \mapsto \underline{x}_k^\top \in \dot{X}_\delta^\top\} \subset p$$

and let $p' < p$. Let Σ' certify p' and let $\mathfrak{C}' = (N', (\lambda'_\eta, X'_\eta : \eta \in K'))$ be the corresponding semantic certificate. By $(\Sigma.4)$, there is some $r \in [\Sigma']^{<\omega} \cap X'_\delta \cap E$. Being certified by Σ' , the following

$$p' \cup r \cup \{\ulcorner \underline{r} \in \dot{X}_\delta \urcorner\}$$

is a condition in $\mathbb{P}_\lambda$ that is stronger than p' . We have shown that following set

$$D_E = \{q \in \mathbb{P}_\lambda : \exists r \in E (r \cup \{\ulcorner \underline{r} \in \dot{X}_\delta \urcorner\} \subset q)\}$$

is dense below p . Let $q \in g \cap D_E$ and let r witness q being an element of D_E .¹⁹ Then we have:

- (i) $r \in E$ by the definition of D_E ,
- (ii) $r \in [\bigcup g]^{<\omega}$ because $r \subset q$ and $q \in [\bigcup g]^{<\omega}$, and
- (iii) $r \in X_\delta$ because $\ulcorner \underline{r} \in \dot{X}_\delta \urcorner \in q \subset \bigcup g$.

Therefore $[\bigcup g]^{<\omega} \cap X_\delta \cap E \neq \emptyset$. □

3.4 Stationary set preservation

Let us write $\mathbb{P} = \mathbb{P}_\kappa$. By Lemma 3.22 and $(\diamond)$, cf. Lemma 3.7, the principle $(\diamond_\mathbb{P})$ also holds in V :

Definition 3.42. $(\diamond_\mathbb{P})$ is the statement that for all $B \subset H_\kappa$, the set

$$\{\lambda \in D : (Q_\lambda; \in, \mathbb{P}_\lambda, A_\lambda) \prec (H_\kappa; \in, \mathbb{P}, B)\}$$

is stationary.²⁰

Lemma 3.43. *Let N be the canonical term model given by a generic filter for $\mathbb{P}$, cf. the proof of Lemma 3.27. Then $\mathbb{P}$ preserves stationary sets of N , i.e. for every $S \subset \omega_1$ that is stationary in N , S is stationary in $V^\mathbb{P}$. In particular, $\mathbb{P}$ preserves stationary subsets of ω_1 in V .*

¹⁹It is easy to see that D_E is definable over $(Q_\lambda; \in, \mathbb{P}_\lambda)$ using parameters in Q_λ .

²⁰We remind the reader that D is from Lemma 3.22.

Proof of Lemma 3.43. Our initial setup is the same as [3]. Let $\bar{p} \in \mathbb{P}$ and $\dot{S}, \dot{C} \in V^{\mathbb{P}}$ and $\alpha_0 < \omega_1$ be such that

- (i) $\bar{p}$ forces “ $\dot{S} \subset \omega_1$ is a stationary set in N which is represented by $[\dot{\alpha}_0]$ ”, and
- (ii) $\bar{p}$ forces “ $\dot{C}$ is club in ω_1 ”.

Fix any $p < \bar{p}$. For $\xi < \omega_1$, let

$$D_\xi = \{q \leq p : \exists \eta \geq \xi (\eta < \omega_1 \wedge q \Vdash \check{\eta} \in \dot{C})\}.$$

Then D_ξ is dense below p for each $\xi < \omega_1$. Also let

$$E = \{(q, \eta) \in \mathbb{P} \times \omega_1 : q \Vdash \check{\eta} \in \dot{C}\},$$

and let

$$\tau = ((D_\xi : \xi < \omega_1), E).$$

We shall identify τ as a subset of H_κ which codes $((D_\xi : \xi < \omega_1), E)$ in a way that ZFC^- is sufficient for the coding/decoding process. By $(\diamond_{\mathbb{P}})$, fix a $\lambda \in D$ such that $p \in \mathbb{P}_\lambda$ and such that

$$(Q_\lambda; \in, \mathbb{P}_\lambda, A_\lambda) \prec (H_\kappa; \in, \mathbb{P}, \tau).$$

Now work in $V^{\text{Col}(\omega, \omega_2^V)}$. Note that both λ and Q_λ are countable in $V^{\text{Col}(\omega, \omega_2^V)}$, so there is a filter $g \ni p$ for $\mathbb{P}_\lambda$ such that g meets every dense subset of $\mathbb{P}_\lambda$ which is definable over the structure $(Q_\lambda; \in, \mathbb{P}_\lambda, A_\lambda)$ using parameters from Q_λ . By Lemma 3.27, $\bigcup g$ is a syntactic quasi-certificate. Let

$$\mathfrak{C} = (N, (\lambda_\delta, X_\delta : \delta \in K))$$

and

$$e : \omega_1 \rightarrow N$$

be the corresponding semantic quasi-certificate and surjection of $\bigcup g$, respectively. We write $S = [\dot{\alpha}_0]$, i.e., the set represented by $\dot{\alpha}_0$ in the term model N .

Claim 3.44. *There is an elementary embedding $\sigma : N \rightarrow N^*$ such that N^* is transitive and $\omega_1 \in \sigma(S)$.*

Proof of Claim 3.44. We have that $p \Vdash_{\mathbb{P}_\lambda} \dot{S} \subset \omega_1$ is a stationary set in N which is represented by $[\dot{\alpha}_0]$ since

$$(Q_\lambda; \in, \mathbb{P}_\lambda, A_\lambda) \prec (H_\kappa; \in, \mathbb{P}, \tau).$$

So that S is a stationary set in N . We put S in a generic filter for $(\text{NS}_{\omega_1}^{+N}; \subset)$ and write $\sigma : N \rightarrow N^*$ for the corresponding generic ultrapower. Then N^* is transitive by the precipitousness of $\text{NS}_{\omega_1}^N$.²¹ That $\omega_1 \in \sigma(S)$ is a standard fact. $\square$ (Claim 3.44)

Let $\mathcal{I} = (M_i, \pi_{ij} : i \leq j \leq \omega_1)$ in N be a generic iteration by the nonstationary ideal and its images with $H_{\omega_2}^V$ being the final iterate.²² Now we apply σ to stretch $\mathcal{I}$; we write $(M_i, \pi_{ij} : i \leq j \leq \nu) = \sigma(\mathcal{I})$ where $\nu = \omega_1^{N^*}$.²³ Since $\text{NS}_{\omega_1}^V$ is saturated in V , we may lift up $\pi_{\omega_1\omega_1+1}$ to act on all of V , then by induction we may lift up the entire iteration $(M_i, \pi_{ij} : \omega_1 \leq i \leq j \leq \mu)$. Write $\pi : V \rightarrow M$, where M is transitive, for the liftup of $\pi_{\omega_1\nu}$. Also see the diagram below.

$$\begin{array}{ccccc} N & \xrightarrow{\sigma} & N^* & & \\ \Psi & & \Psi & & \\ M_0 & \xrightarrow{\pi_{0\omega_1}} & H_{\omega_2}^V & \xrightarrow{\pi_{\omega_1\nu}} & M_\nu \\ \cap & & \cap & & \\ V & \xrightarrow{\pi} & M & & \end{array}$$

Claim 3.45. *The following hold.*

$$(1) \sigma \restriction H_{\omega_2}^V = \pi \restriction H_{\omega_2}^V.$$

$$(2) \sigma(X) = \pi(X) \text{ for any } X \subset H_{\omega_2}^V \text{ that is an element of } V \cap N.$$

Proof of Claim 3.45. We first prove (1). Fix an $x \in H_{\omega_2}^V$. Let $\bar{x}$ and $i < \omega_1$ be such that $x = \pi_{i\omega_1}(\bar{x})$. Note that $\sigma(\bar{x}) = \bar{x}$ since $\bar{x}$ is hereditarily countable in N .

²¹ $\text{NS}_{\omega_1}^N$ is precipitous because N witnesses the weak honest consistency of $\varphi(A)$, cf. Definition 2.11.

²²The existence of such an $\mathcal{I}$ is given by Lemma 2.6.

²³It is easily verifiable that the initial segment of length $\omega_1 + 1$ of $\sigma(\mathcal{I})$ is $\mathcal{I}$.

Now we have

$$\begin{aligned}
 \pi(x) &= \pi_{\omega_1\nu}(x) = \pi_{\omega_1\nu}(\pi_{i\omega_1}(\bar{x})) \\
 &= \pi_{i\nu}(\bar{x}) \\
 &= \sigma(\pi_{i\omega_1})(\bar{x}) \\
 &= \sigma(\pi_{i\omega_1})(\sigma(\bar{x})) \\
 &= \sigma(\pi_{i\omega_1}(\bar{x})) = \sigma(x).
 \end{aligned}$$

We prove (2). We assume without loss of generality that X is a set of ordinals. Since ω_2^V has ω -cofinality in N , we may let (in N)

$$X = \bigcup \{X_n : n < \omega\},$$

where $X_n \in H_{\omega_2}^V$ for each $n < \omega$. We have that

$$\begin{aligned}
 \sigma(X) &= \bigcup \sigma(\{X_n : n < \omega\}) \\
 &= \bigcup \{\sigma(X_n) : n < \omega\} \\
 &= \bigcup \{\pi(X_n) : n < \omega\} \\
 &= \pi(X).
 \end{aligned}$$

The first two equalities are trivial. The third one is given by (1). As for the last equality, we show that π is continuous at ω_2^V , i.e., $\pi(\omega_2^V) = \bigcup \pi \text{``}\omega_2^V$. To see this, note that $\pi_{\omega_1\nu}$ is a cofinal embedding and the ordinal height of M_ν is $\sigma(\omega_2^V)$ which has ω -cofinality in N^* , so that

$$\sigma(\omega_2^V) \geq \pi(\omega_2^V) \geq \bigcup \pi \text{``}\omega_2^V = \bigcup \sigma \text{``}\omega_2^V = \sigma(\omega_2^V),$$

hence $\pi(\omega_2^V) = \bigcup \pi \text{``}\omega_2^V$. Then it is easy to prove the last equality. $\square$ (Claim 3.45)

Now we can proceed with the arguments from the original paper [3]. Our goal is to show that $\pi \text{``}\bigcup g$ can be extended to a $\pi(\kappa)$ -syntactic certificate Σ for $\pi(p)$.

We define

$$\begin{aligned}
 K^* &= K \cup \{\omega_1\} \\
 \lambda_\delta^* &= \pi(\lambda_\delta), \text{ for } \delta \in K \\
 \lambda_{\omega_1}^* &= \pi(\lambda) \\
 X_\delta^* &= \pi^{\text{``}}X_\delta, \text{ for } \delta \in K \\
 X_{\omega_1}^* &= \pi^{\text{``}}Q_\lambda \\
 \mathfrak{C}^* &= (N^*, (\lambda_\delta^*, X_\delta^* : \delta \in K^*)).
 \end{aligned}$$

Claim 3.46. $\mathfrak{C}^*$ is a $\pi(\kappa)$ -semantic certificate relative to $\pi(A)$, $\pi(H_{\omega_2}^V)$, $\pi((A_\mu : \mu \in C \cup \{\kappa\}))$, $\pi((Q_\mu : \mu \in C \cup \{\kappa\}))$, and $\pi((\mathbb{P}_\mu : \mu \in C \cup \{\kappa\}))$.²⁴

Proof of Claim 3.46. Conditions $(\mathfrak{C}.1)$, $(\mathfrak{C}.3) - (\mathfrak{C}.5)$ follow directly from the elementarity of π and the definitions of the *-superscripted objects and we shall omit the details. As for $(\mathfrak{C}.2)$, just notice that N^* is countable in $V^{\text{Col}(\omega, \omega_2^V)}$ whereas $\pi(\kappa)$ is uncountable. This of course means that the ordinal height of N^* is less than $\pi(\kappa)$. $\square$ (Claim 3.46)

Let $\tilde{e} : \omega_1 \rightarrow N^*$ be the partial function such that if $e(\alpha) \in H_{\omega_2}^V \cup \{A, \omega_2^V, H_{\omega_2}^V\}$, then $\tilde{e}(\alpha) = \pi(e(\alpha))$. Then by Claim 3.46, the elementarity of π , and the fact that $\bigcup g$ is a syntactic quasi-certificate, it is easily shown that for any $\mathcal{L}$ -formula Φ , for any $\alpha_1, \dots, \alpha_k \in \text{dom}(\tilde{e})$, and for any $x_1, \dots, x_\ell \in H_{\omega_2}^V \cup \{A, \omega_2^V, H_{\omega_2}^V\}$,

$$\ulcorner \dot{N} \models \Phi(\dot{\alpha}_1, \dots, \dot{\alpha}_k, \underline{\pi(x_1)}, \dots, \underline{\pi(x_\ell)}) \urcorner \in \pi^{\text{``}} \bigcup g$$

iff

$$N^* \models \Phi(\tilde{e}(\alpha_1), \dots, \tilde{e}(\alpha_k), \pi(x_1), \dots, \pi(x_\ell)).$$

Similarly, by the construction of the *-superscripted objects, $\pi^{\text{``}} \bigcup g$  also satisfies its own versions of  $(\Sigma.2)$  and  $(\Sigma.3)$ . We verify that  $\pi^{\text{``}} \bigcup g$ satisfies $(\Sigma.4)$. Note that $[\pi^{\text{``}} \bigcup g]^{<\omega} = \pi^{\text{``}} g$.

²⁴Semantic certificates in the present thesis are not necessarily certifiable by a syntactic certificate. This is a main difference in terms of naming conventions compared with [3]. We have yet to prove that $\mathfrak{C}^*$ as in the claim is indeed certifiable. See the second remark after Definition 3.11.

Claim 3.47. $\pi \text{“} g \cap X_\delta^* \cap E \neq \emptyset \text{ for every } \delta \in K^* \text{ and every } E \subset \pi(\mathbb{P})_{\lambda_\delta^*} \text{ which is dense in } \pi(\mathbb{P})_{\lambda_\delta^*} \text{ and definable over the structure } (\pi(Q)_{\lambda_\delta^*}; \in, \pi(\mathbb{P})_{\lambda_\delta^*}, \pi(A)_{\lambda_\delta^*}) \text{ using parameters in } X_\delta^* \text{”}$.²⁵

Proof of 3.47. Case 1. $\delta \in K$. We have $X_\delta^* = \pi \text{“} X_\delta$ and $\pi \text{“} g \cap X_\delta^* = \pi \text{“} (g \cap X_\delta)$. In this case we also have

$$(\pi(Q)_{\lambda_\delta^*}; \in, \pi(\mathbb{P})_{\lambda_\delta^*}, \pi(A)_{\lambda_\delta^*}) = (\pi(Q_{\lambda_\delta}); \in, \pi(\mathbb{P}_{\lambda_\delta}), \pi(A_{\lambda_\delta})).$$

Recall that $\bigcup g$ is a syntactic quasi-certificate. Then by $(\Sigma.4)$, $g \cap X_\delta \cap E \neq \emptyset$ for every E which is dense in $\mathbb{P}_{\lambda_\delta}$ and definable over the structure $(Q_{\lambda_\delta}; \in, \mathbb{P}_{\lambda_\delta}, A_{\lambda_\delta})$ using parameters in X_δ . Now the above statement can be directly pushed up by π . The claim follows.

Case 2. $\delta = \omega_1$. We note that $X_{\omega_1}^* = \pi \text{“} Q_\lambda$ and

$$(\pi(Q)_{\lambda_{\omega_1}^*}; \in, \pi(\mathbb{P})_{\lambda_{\omega_1}^*}, \pi(A)_{\lambda_{\omega_1}^*}) = (\pi(Q_\lambda); \in, \pi(\mathbb{P}_\lambda), \pi(A_\lambda)).$$

The rest is essentially the same as Case 1. □ (Claim 3.47)

Now we arbitrarily expand the partial function $\tilde{e}$ to a surjection $e^* : \omega_1^{N^*} \rightarrow N^*$ and derive in the natural way a collection $\Sigma \supset \pi \text{“} \bigcup g$ of $\mathcal{L}$ -formulas. By Claim 3.47 and the discussion above it, it is easy to see that this Σ is a $\pi(\kappa)$ -syntactic certificate with $\mathfrak{C}^*$ and e^* being the corresponding semantic certificate and surjection, respectively. It is also easy to see that

$$q^* = \pi(p) \cup \{ \ulcorner \dot{N} \models \underline{\omega}_1 \in \pi(\dot{\alpha}_0) = \dot{\alpha}_0 \urcorner, \ulcorner \underline{\omega}_1 \mapsto \underline{\pi(\lambda)} \urcorner \}$$

is a condition in $\pi(\mathbb{P})$ as being certified by Σ . By the elementarity of π , there is some $\delta < \omega_1$ such that

$$q = p \cup \{ \ulcorner \dot{N} \models \underline{\delta} \in \dot{\alpha}_0 \urcorner, \ulcorner \underline{\delta} \mapsto \underline{\lambda} \urcorner \}$$

is a condition in $\mathbb{P}$. Clearly q forces “ $\dot{S} \subset \omega_1$ is a stationary set represented by

²⁵In the statement of this claim, $\pi(\mathbb{P})_{\lambda_\delta^*}$ should really be understood as the λ_δ^* -th component of the sequence of forcings $\pi((\mathbb{P}_\mu : \mu \in C \cup \{\kappa\}))$. Same applies to $\pi(A)_{\lambda_\delta^*}$ and $\pi(Q)_{\lambda_\delta^*}$.

$[\dot{\alpha}_0]$ in N'' as q is a stronger condition than p ; then

$$q \Vdash \check{\delta} \in \dot{S}$$

since $\ulcorner \dot{N} \models \underline{\delta} \in \dot{\alpha}_0 \urcorner \in q$. We remain to show that $q \Vdash \check{\delta} \in \dot{C}$.

Claim 3.48. $q \Vdash \text{“}\check{\delta} \text{ is a limit point of } \dot{C}\text{”}$. In particular, $q \Vdash \check{\delta} \in \dot{C}$.

Proof of Claim 3.48. The argument is the same as [3, Claim 3.17]. Suppose the claim is false. Let us assume that there is some $r \in \mathbb{P}$ which is stronger q such that for some $\eta < \delta$,

$$r \Vdash \dot{C} \cap \check{\delta} \subset \check{\eta}.$$

Let Σ' certify r , and let

$$\mathfrak{C}' = (N', (\lambda'_\xi, X'_\xi : \xi \in K'))$$

be the corresponding semantic certificate. Since $\eta < \delta$, by $(\mathfrak{C}.5)$ we have that $D_\eta \cap Q_\lambda$ is definable over $(Q_\lambda; \in, \mathbb{P}_\lambda, A_\lambda)$ using parameters from X'_δ ; then by $(\Diamond_{\mathbb{P}})$,

$$(Q_\lambda; \in, \mathbb{P}_\lambda, A_\lambda) \prec (H_\kappa; \in, \mathbb{P}, \tau)$$

so that $D_\eta \cap Q_\lambda$ is dense in $\mathbb{P}_\lambda$. Applying $(\Sigma.4)$ we pick some $s \in [\Sigma']^{<\omega} \cap X'_\delta \cap D_\eta$, and let $\eta' \geq \eta$ be the least such that $s \Vdash \check{\eta}' \in \dot{C}$.²⁶ This η' is definable so $\eta' \in X'_\delta$. And clearly $\eta' < \delta$ by $(\mathfrak{C}.5)$. Hence $s \Vdash \dot{C} \cap \check{\delta} \not\subset \check{\eta}$ and s is incompatible with r . But this contradicts the fact that both s and r are certified by Σ' .

□ (Claim 3.48, Lemma 3.43)

We finish the proof of Theorem 3.1 by verifying $V^{\mathbb{P}} \models \varphi(A)$.

Lemma 3.49. $V^{\mathbb{P}} \models \varphi(A)$.

Proof. Let us write N for the canonical term model given by a generic filter for $\mathbb{P}$. By Lemma 3.27, $N \models \varphi(A)$. Note that $\varphi(A)$ is a Σ_1 -formula in $\mathcal{L}_{\dot{\in}, \text{NS}_{\omega_1}}$, hence it holds true in any stationary set preserving extension of N . By Lemma 3.43, $V^{\mathbb{P}} \models \varphi(A)$. □

²⁶Recall $D_\eta = \{q \leq p : \exists \eta' \geq \eta (\eta' < \omega_1 \wedge q \Vdash \check{\eta}' \in \dot{C})\}$.

Chapter 4

The Π_1 -Nondefinability of NS_{ω_1}

Theorem 4.1. *Assume BMM and that NS_{ω_1} is saturated. Then NS_{ω_1} is not boldface Π_1 -definable over H_{ω_2} .*

Corollary 4.2. *Assume MM. Then NS_{ω_1} is not boldface Π_1 -definable over H_{ω_2} .*

Proof. MM trivially implies BMM. That MM implies the saturation of NS_{ω_1} was shown in [9], see also [17, Theorem 37.16]. $\square$

Proof of Theorem 4.1. Suppose that Theorem 4.1 is false, then there exist Σ_1 -formula φ_{stat} and $X \subset \omega_1$ such that for all $S \subset \omega_1$,

$$S \text{ is stationary} \iff \varphi_{\text{stat}}(S, X).$$

We aim to derive a contradiction. Recall that by Todorcevic [25], BMM implies $2^{\aleph_1} = \aleph_2$.²⁷

Lemma 4.3. *In $V^{\text{Col}(\omega, \omega_2)}$, there is a transitive stationary set preserving extension N of $H_{\omega_2}^V$ such that*

(i) $N \models \exists D \subset \omega_1 (\varphi_{\text{stat}}(D, X) \wedge \forall S \in \text{NS}_{\omega_1}^{+V}, S \setminus D \text{ is stationary}),$

(ii) $N \models \text{cf } \omega_2^V = \omega \wedge H_{\omega_2}^V \text{ has size } \aleph_1,$

(iii) $N \models \text{“NS}_{\omega_1} \text{ is precipitous”}.$

Proof. Fix a sufficiently large θ . Let $j : M \rightarrow H_\theta$ be such that

²⁷In fact, Moore [22] shows that BPFA already implies $2^{\aleph_0} = \aleph_2$. But Todorcevic’s result is all we need here.

(a) M is a transitive set of size $\aleph_1$,

(b) $\text{crit}(j) = \omega_2^M$, and

(c) $\text{cf } \omega_2^M = \omega$.

Clearly both $H_{\omega_2}^M$ and $\text{NS}_{\omega_1}^{+M}$ have size $\aleph_1$. Since BMM implies MA_{ω_1} which in turn implies the nonstationary ideal is not ω_1 -dense, there is a stationary $D \subset \omega_1$ such that

$$\forall S \in \text{NS}_{\omega_1}^{+M}, S \setminus D \text{ is stationary.}$$

Now we have that H_θ is a stationary set preserving extension of $H_{\omega_2}^M$ such that

(M.i) $H_\theta \models \exists D \subset \omega_1 (\varphi_{\text{stat}}(D, X) \wedge \forall S \in \text{NS}_{\omega_1}^{+M}, S \setminus D \text{ is stationary}),$

(M.ii) $H_\theta \models \text{cf } \omega_2^M = \omega \wedge H_{\omega_2}^M \text{ has size } \aleph_1,$

(M.iii) $H_\theta \models \text{“NS}_{\omega_1} \text{ is precipitous”}.$

By Shoenfield absoluteness, there is an $\bar{N} \in M^{\text{Col}(\omega, \omega_2^M)}$ such that the above items hold. Now apply the elementarity of j to the above items (M.i) - (M.iii), there is an $N \in V^{\text{Col}(\omega, \omega_2)}$ such that

(V.i) $N \models \exists D \subset \omega_1 (\varphi_{\text{stat}}(D, X) \wedge \forall S \in \text{NS}_{\omega_1}^{+V}, S \setminus D \text{ is stationary}),$

(V.ii) $N \models \text{cf } \omega_2^V = \omega \wedge H_{\omega_2}^V \text{ has size } \aleph_1,$

(V.iii) $N \models \text{“NS}_{\omega_1} \text{ is precipitous”}.$ □ (Lemma 4.3)

Consider the following statement:

$$\varphi(\text{NS}_{\omega_1}^{+V}) \equiv \exists D \subset \omega_1 (\varphi_{\text{stat}}(D, X) \wedge \forall S \in \text{NS}_{\omega_1}^{+V}, S \setminus D \text{ is stationary})$$

By Lemma 4.3, $\varphi(\text{NS}_{\omega_1}^{+V})$ is weakly honestly consistent, cf. Definition 2.11. So by Theorem 3.1, there is a stationary set preserving forcing $\mathbb{P}$ such that

$$V^{\mathbb{P}} \models \varphi(\text{NS}_{\omega_1}^{+V}).^{28}$$

²⁸We treat $\text{NS}_{\omega_1}^{+V}$ as a fixed parameter.

Now fix a $D \subset \omega_1$ in $V^{\mathbb{P}}$ which witnesses the existential quantifier of $\varphi(\text{NS}_{\omega_1}^{+V})$ and let $\mathbb{S} \in V^{\mathbb{P}}$ be the standard forcing that shoots a club through $\omega_1 \setminus D$,²⁹ namely, the conditions of $\mathbb{S}$ are closed and bounded subsets of $\omega_1 \setminus D$, and the order is given by end-extension. Even though $\mathbb{S}$ destroys the stationarity of D , it preserves stationary sets in V .

Lemma 4.4. $\mathbb{P} * \mathbb{S}$ is stationary set preserving.

Proof. Let $S \subset \omega_1$ be a stationary set in V . Let g be a filter for $\mathbb{P}$ which is generic over V . Let $p \in \mathbb{S}$ and $\dot{C} \in V[g]^{\mathbb{S}}$ be such that

$$p \Vdash_{V[g]} \dot{C} \subset \omega_1 \text{ is club.}$$

We aim to find some $q < p$ such that

$$q \Vdash_{V[g]} \dot{C} \cap \check{S} \neq \emptyset.$$

Work in $V[g]$. Let θ be sufficiently large, and let $X \prec H_{\theta}^{V[g]}$ be such that

- (a) X is countable,
- (b) $\{\mathbb{S}, p, \dot{C}\} \subset X$, and
- (c) $X \cap \omega_1 = \alpha$ for some $\alpha \in S \setminus D$.

The above item (c) is possible since $S \setminus D$ is stationary in $V[g]$. For $\xi < \alpha$, let

$$D_{\xi} = \{q \in \mathbb{S} : \exists \eta \geq \xi (q \Vdash \check{\eta} \in \dot{C})\}.$$

These D_{ξ} 's are elements of X and dense below p , so that there is a filter $h \in V[g]$ for $\mathbb{S}$ such that $p \in h$ and $h \cap D_{\xi} \cap X \neq \emptyset$ for all $\xi < \alpha$. Let

$$q = (\bigcup h) \cup \{\alpha\}.$$

Claim 4.5. $q \Vdash \check{\alpha} \in \dot{C}$.

Proof of Claim 4.5. Since $h \cap D_{\xi} \cap X \neq \emptyset$ for each $\xi < \alpha$, it is easy to see that for each $\xi < \alpha$, there are η_{ξ} and α_{ξ} such that

²⁹Notice that $\omega_1 \setminus D$ is stationary.

- (a) $\xi \leq \eta_\xi < \alpha$,
- (b) $\alpha_\xi < \alpha$, and
- (c) $q \restriction (\alpha_\xi + 1) \Vdash \check{\eta}_\xi \in \dot{C}$.

This means that q forces into $\dot{C}$ an unbounded subset of α , i.e. the set of all η_ξ 's. Clearly q also forces α into $\dot{C}$ since q is stronger than p . $\square$ (Claim 4.5)

But $\check{\alpha} \in \check{S}$ is forced by every condition in $\mathbb{S}$ since α is already an element of S . This means that $q \Vdash \check{\alpha} \in \dot{C} \cap \check{S}$. $\square$ (Lemma 4.4)

Recall that $D \subset \omega_1$ is a stationary set added by $\mathbb{P}$ such that $V^{\mathbb{P}} \models \varphi_{\text{stat}}(D, X)$. As the formula $\varphi_{\text{stat}}(D, X)$ is Σ_1 , by upward absoluteness, it still holds in $V^{\mathbb{P} * \mathbb{S}}$. However, the stationarity of D is destroyed by the club shooting forcing $\mathbb{S}$. This means that

$$V^{\mathbb{P} * \mathbb{S}} \models \exists D (\varphi_{\text{stat}}(D, X) \wedge D \text{ is nonstationary}).^{30}$$

By Lemma 4.4, $\mathbb{P} * \mathbb{S}$ is stationary set preserving so we can use BMM to pull back the above statement to V :

$$V \models \exists D (\varphi_{\text{stat}}(D, X) \wedge D \text{ is nonstationary}).$$

But this contradicts our assumption that

$$\varphi_{\text{stat}}(D, X) \iff D \text{ is stationary}$$

in V . We have thus finished the proof of Theorem 4.1. $\square$ (Theorem 4.1)

4.1 Generalized descriptive set theoretic results

We study generalized descriptive set theoretic aspects of NS_{ω_1} in this section. Let κ be a regular uncountable cardinal. Write $\mathcal{N}_\kappa = {}^\kappa\kappa$. We sometimes confuse subsets of κ with their characteristic functions.

Definition 4.6. We write $O_s = \{f \in \mathcal{N}_\kappa : s \subset f\}$ for $s \in {}^{<\kappa}\kappa$. Let $1 \leq k < \omega$. Let $X \subset \mathcal{N}_\kappa^k$. Then

³⁰This statement is Σ_1 in the language $\mathcal{L}_\in$ without using the predicate for NS_{ω_1} so that in Theorem 4.1 we only need BMM instead of BMM^{++} .

- X is *open* if $X = \bigcup \{ \prod_{i < k} O_{s_i} : (s_i : i < k) \in I \}$ for some $I \subset (<^\kappa \kappa)^k$,
- X is *closed* if $\mathcal{N}_\kappa^k \setminus X$ is open,
- X is Σ_1^1 if X is the projection of some closed set $Y \subset \mathcal{N}_\kappa^{k+1}$, i.e.,

$$X = \{ (f_0, \dots, f_{k-1}) : \exists g (f_0, \dots, f_{k-1}, g) \in Y \},$$

- X is Π_1^1 if $\mathcal{N}_\kappa^k \setminus X$ is Σ_1^1 .

We mainly focus on the case when $\kappa = \omega_1$. The following is a standard fact that a subset of $\mathcal{N}_{\omega_1}^k$ being closed can be characterized in terms of trees.

Lemma 4.7. *Let $1 \leq k < \omega$. Let $X \subset \mathcal{N}_{\omega_1}^k$. The following are equivalent:*

- (1) X is closed,
- (2) $X = [U]$ for some tree $U \subset (<^{\omega_1} \omega_1)^k$, where $[U]$ is the collection of branches of U , i.e.,

$$[U] = \{ (f_0, \dots, f_{k-1}) : \forall \alpha < \omega_1 (f_0 \restriction \alpha, \dots, f_{k-1} \restriction \alpha) \in U \}.$$

Proof. [23, Lemma 7.1] gives a proof of this result for the Baire space $\mathcal{N} = {}^\omega \omega$. That proof also works for generalized Baire spaces. $\square$

Corollary 4.8. *Let $1 \leq k < \omega$. Let $X \subset \mathcal{N}_{\omega_1}^k$. The following are equivalent:*

- (1) X is Σ_1^1 ,
- (2) $X = p[U]$ for some tree $U \subset (<^{\omega_1} \omega_1)^{k+1}$, where $p[U]$ is the projection of $[U]$, i.e.,

$$p[U] = \{ (f_0, \dots, f_{k-1}) : \exists g (f_0, \dots, f_{k-1}, g) \in [U] \}.$$

Proof. Immediate from the definition of Σ_1^1 sets and Lemma 4.7. $\square$

We have another standard fact:

Lemma 4.9. NS_{ω_1} is Σ_1^1 .

Proof. $\text{NS}_{\omega_1} = p[U]$ where U is the tree consisting of all pairs $(s, t) \in (<^{\omega_1} \omega_1)^2$ such that

- $\text{dom}(s) = \text{dom}(t)$,
- $\text{ran}(s) \cup \text{ran}(t) \subset \{0, 1\}$,
- for all $\xi \in \text{dom}(s)$, if $s(\xi) = 1$ then $t(\xi) = 0$,
- t is closed.³¹

□

4.1.1 Every set is Σ_1^1 under MA_{ω_1}

In the light of Theorem 4.1, it is natural to conjecture that NS_{ω_1} is not Π_1^1 under similar assumptions. As a straightforward application of Theorem 4.10, Corollary 4.12 shows that this is completely not the case, however.

Theorem 4.10. *Let κ be a regular uncountable cardinal. Assume that $2^\lambda = 2^\kappa$ for some $\lambda < \kappa$. Let $1 \leq \ell < \omega$. Let $X \subset \mathcal{N}_\kappa^\ell$. Then X is Σ_1^1 .*³²

Proof. Without loss of generality let us assume that $\ell = 1$.³³ For $x \in {}^\lambda 2$, we write $x \smallfrown 0 \in {}^\kappa 2 \subset {}^\kappa \kappa$ for the sequence such that

$$x \smallfrown 0(\xi) = \begin{cases} x(\xi) & \text{if } \xi < \lambda, \\ 0 & \text{if } \lambda \leq \xi < \kappa. \end{cases}$$

Let $f \mapsto y_f$ be a bijection from ${}^\kappa \kappa$ to ${}^\lambda 2$. We only need to show that the following set

$$Y = \{(f, y_f \smallfrown 0) : f \in X\}$$

is closed because then

$$X = \{f : \exists g (f, g) \in Y\}$$

is Σ_1^1 as it is the projection of Y . Now let U be the tree consisting of pairs $(s, t) \in ({}^{<\kappa} \kappa)^2$ such that

- $\text{dom}(s) = \text{dom}(t)$,

³¹In the sense that for all increasing $\{\xi_n : n < \omega\} \subset \text{dom}(t)$ with the property that $t(\xi_n) = 1$ for every $n < \omega$, if $\sup_{n < \omega} \xi_n = \xi < \text{dom}(t)$, then $t(\xi) = 1$.

³²It is clear from the proof that the theorem also holds for generalized Cantor spaces $({}^\kappa 2)^\ell$.

³³For the general case, we simply replace the bijection from ${}^\kappa \kappa$ to ${}^\lambda 2$ in what follows with a bijection from $({}^\kappa \kappa)^\ell$ to ${}^\lambda 2$ and make obvious notational modifications accordingly.

- there is an $f \in X$ such that
 - (1) $t \restriction \lambda \subset y_f$,
 - (2) $t(\xi) = 0$ for all $\xi \in \text{dom}(t) \setminus \lambda$, and
 - (3) $s(\xi) = f(\xi)$ for all $\xi \in \text{dom}(s)$.

Claim 4.11. $Y = [U]$.

Proof of Claim 4.11. By the definition of U , it is straightforward to verify that $(f, y_f \restriction 0) \in [U]$ whenever $f \in X$. On the other hand, suppose that $(f, g) \in [U]$. Let $f_\alpha \in X$ witness $(f \restriction \alpha, g \restriction \alpha) \in U$ for $\alpha < \kappa$. Then for all $\alpha, \beta \in \kappa \setminus \lambda$,

$$y_{f_\alpha} = (g \restriction \alpha) \restriction \lambda = g \restriction \lambda = (g \restriction \beta) \restriction \lambda = y_{f_\beta}.$$

Since $f \mapsto y_f$ is a bijection, necessarily we have

$$f_\alpha = f_\beta$$

for all $\alpha, \beta \in \kappa \setminus \lambda$. It is easy to see that for some/all $\alpha \in \kappa \setminus \lambda$, $f = f_\alpha \in X$ and $g = y_f \restriction 0$. Therefore $(f, g) \in Y$. $\square$ (Claim 4.11)

Claim 4.11 shows that Y is closed and hence X is Σ_1^1 . $\square$ (Theorem 4.10)

Corollary 4.12. Assume MA_{ω_1} . Then every subset of $\mathcal{N}_{\omega_1}$ is Σ_1^1 . In particular, $\text{NS}_{\omega_1}^+$ is Σ_1^1 and hence NS_{ω_1} is Π_1^1 .

Proof. MA_{ω_1} implies $2^{\aleph_0} = 2^{\aleph_1}$. Apply Theorem 4.10. $\square$

4.1.2 NS_{ω_1} is not $\text{uB-}\Pi_1^1$

Definition 4.13. Fix a canonical way to code any finite tuple of countable sequences of countable ordinals into a real. Let $1 \leq k < \omega$. We write $x_{\vec{s}}$ for the real which codes $\vec{s} \in (<\omega_1 \omega_1)^k$. Let $C \subset (<\omega_1 \omega_1)^k$. We say that $A \subset \mathbb{R}$ codes C if

$$A = \{x_{\vec{s}} : \vec{s} \in C\}.$$

Definition 4.14. Let $1 \leq k < \omega$. We say a set $C \subset (<\omega_1 \omega_1)^k$ is *universally Baire in the codes* if the set of reals which codes C , i.e. the set $\{x_{\vec{s}} : \vec{s} \in C\}$, is universally Baire.

Definition 4.15 (The uB-projective hierarchy). Let $1 \leq k < \omega$. Let $X \subset \mathcal{N}_{\omega_1}^k$. Then

- X is uB-open if $X = \bigcup \{ \prod_{i < k} O_{s_i} : (s_i : i < k) \in C \}$ for some $C \subset ({}^{<\omega_1}\omega_1)^k$ which is universally Baire in the codes,³⁴
- X is uB-closed if $\mathcal{N}_{\omega_1}^k \setminus X$ is uB-open,
- X is uB- Σ_1^1 if X is the projection of some uB-closed set $Y \subset \mathcal{N}_{\omega_1}^{k+1}$,
- X is uB- Π_1^1 if $\mathcal{N}_{\omega_1}^k \setminus X$ is uB- Σ_1^1 ,

and for $1 \leq n < \omega$, we inductively define

- X is uB- Σ_{n+1}^1 if X is the projection of some uB- Π_n^1 set $Y \subset \mathcal{N}_{\omega_1}^{k+1}$, and
- X is uB- Π_{n+1}^1 if $\mathcal{N}_{\omega_1}^k \setminus X$ is uB- Σ_{n+1}^1 .

Theorem 4.16. Assume Γ^∞ -BMM and that there is a Woodin cardinal. Then NS_{ω_1} is not uB- Π_1^1 .

Proof. The proof is in essence the same as [15, Theorem 2.1]. Suppose the theorem is false, then there are (class-sized) trees T, U witnessing the universal Baire-ness of some set of reals such that for all $S \subset \omega_1$,

$$S \text{ is stationary} \iff \varphi_{\text{uB}}(S, p[U]) \equiv \exists X \in \mathcal{N}_{\omega_1} \forall \xi < \omega_1 (x_{(S \restriction \xi, X \restriction \xi)} \in p[U]).$$

Let δ be a Woodin cardinal and let g be a generic filter for the stationary tower $\mathbb{P}_{<\delta}$ such that the generic ultrapower embedding

$$\sigma : V \rightarrow M \subset V[g]$$

has critical point ω_2^V . It is easy to see that $\text{NS}_{\omega_1}^{+V}$ has size $\aleph_1$ in M .

Claim 4.17. $p[\sigma(U)]^M = p[\sigma(U)]^{V[g]} = p[U]^{V[g]} = p[U]^M$.

³⁴It is worth noting that the collection of all uB-open sets does not form a topology. However, if there is a proper class of Woodin cardinals and let $A \subset \mathbb{R}$ be universally Baire, then from the point of view of $L(A, \mathbb{R})$, the collection of all uB-open sets forms a topology as it coincides with the open sets as in Definition 4.6. This situation around the uB-projective hierarchy might share similarities with the so-called “pointless topology” in category theory.

Proof of Claim 4.17. The first and third equalities hold because

$${}^{<\delta}M \cap V[g] \subset M,$$

cf. Fact 2.32. We prove the second equality. Suppose that $f : \omega \rightarrow \text{Ord}$ witnesses $x \in p[U]^M$. Then $\tilde{f} : \omega \rightarrow \text{Ord}$ defined as $\tilde{f}(n) = \sigma(f(n))$ is a function in M and witnesses $x \in p[\sigma(U)]^M$.³⁵ This shows that

$$p[U]^M \subset p[\sigma(U)]^M.$$

On the other hand, the above inclusion cannot be proper since by universal Baire-ness and elementarity,

$$p[U]^M = p[U]^{V[g]} = \mathbb{R}^{V[g]} \setminus p[T]^{V[g]} = \mathbb{R}^M \setminus p[T]^M,^{36}$$

and

$$p[\sigma(U)]^M = \mathbb{R}^M \setminus p[\sigma(T)]^M.$$

So the second equality in the claim also holds. $\square$ (Claim 4.17)

Now the elementarity of σ , $(|\text{NS}_{\omega_1}^{+V}| = \aleph_1)^M$, and $(\text{NS}_{\omega_1} \text{ is not } \omega_1\text{-dense})^M$ together give us

$$M \models \exists D \subset \omega_1 (\varphi_{\text{uB}}(D, p[\sigma(U)]) \wedge \forall S \in \text{NS}_{\omega_1}^{+V} (S \setminus D \text{ is stationary})),$$

and then by Claim 4.17 and the fact that ${}^{<\delta}M \cap V[g] \subset M$,

$$V[g] \models \exists D \subset \omega_1 (\varphi_{\text{uB}}(D, p[U]) \wedge \forall S \in \text{NS}_{\omega_1}^{+V} (S \setminus D \text{ is stationary})).$$

Same as before, fix a D such that the above is satisfied and let h be a generic filter for forcing which shoots a club through the complement of D , we have

$$V[g * h] \models \varphi_{\text{uB}}(D, p[U]) \wedge D \text{ is nonstationary.}$$

³⁵Here $\tilde{f} \in M$ is given by ${}^{<\delta}M \cap V[g] \subset M$.

³⁶Notice that $\mathbb{R}^{V[g]} = \mathbb{R}^M$; again, this follows ${}^{<\delta}M \cap V[g] \subset M$.

Since $V[g * h]$ preserves stationary sets of V , by Γ^∞ -BMM,

$$V \models \exists D (\varphi_{\text{uB}}(D, p[U]) \wedge D \text{ is nonstationary}).$$

But this contradicts our assumption. $\square$ (Theorem 4.16)

At the time of writing the present thesis, it is unknown if the existence of a Woodin cardinal as in the above theorem can be relaxed to weaker assumptions or replaced by “ NS_{ω_1} is saturated”. We now proceed to proving some basic properties of sets appearing in the uB-projective hierarchy. Checking through the definitions, we have for every $n < \omega$, $\text{uB-}\Sigma_n^1 \subset \Sigma_n^1$ and $\text{uB-}\Pi_n^1 \subset \Pi_n^1$. Thus the (1) $\implies$ (2) direction as in Lemma 4.7 also holds for uB-closed sets:

Lemma 4.18. *Let $1 \leq k < \omega$. Let $X \subset \mathcal{N}_{\omega_1}^k$. Suppose that X is uB-closed, then there is a tree $U \subset (<_{\omega_1} \omega_1)^k$ such that $X = [U]$.*

The inverse of the above lemma is not true as in general we cannot recover a universally Baire code set merely from a tree. With additional assumptions, we are able to prove a slightly stronger version of Lemma 4.18 in the sense that the tree representing a uB-closed set can be chosen in a determinacy model of the form $L(A, \mathbb{R})$, where $A \subset \mathbb{R}$ is universally Baire.

Lemma 4.19. *Assume Γ^∞ -BMM $^{++}$ and that there is a proper class of Woodin cardinals. Let $1 \leq k < \omega$ and let $X \subset \mathcal{N}_{\omega_1}^k$ be uB-closed. Suppose that*

$$X = \mathcal{N}_{\omega_1}^k \setminus \cup \{ \prod_{i < k} O_{s_i} : (s_i : i < k) \in C \}$$

where $C \subset (<_{\omega_1} \omega_1)^k$ is universally Baire in the codes. Let $A \subset \mathbb{R}$ be the universally Baire set which codes C , (cf. Definition 4.13). Then there is a tree $U \subset (<_{\omega_1} \omega_1)^k$ in $L(A, \mathbb{R})$ such that $X = [U]$. Moreover, the tree U as above is also universally Baire in the codes.

Proof. Without loss of generality let us assume that $k = 1$. Notice that $L(A, \mathbb{R})$ is a models of AD^+ and $C \in L(A, \mathbb{R})$. Let

$$\dot{X} = \{ (\dot{f}, p) : p \Vdash_{\mathbb{P}_{\max}}^{L(A, \mathbb{R})} \forall s \in C (s \not\subset \dot{f}) \}$$

and let

$$U = \{ s \in <_{\omega_1} \omega_1 : \Vdash_{\mathbb{P}_{\max}}^{L(A, \mathbb{R})} \exists f \in \dot{X} (s \subset f) \}.$$

Then $U \in L(A, \mathbb{R})$ by the definability of the forcing relation. By Asperó–Schindler’s result, cf. Theorem 2.27, $(*)_{\Gamma^\infty}$ holds true, so that $\mathcal{P}(\omega_1)$ is contained in the $\mathbb{P}_{\max}$ extension of $L(A, \mathbb{R})$. Therefore

$$[U]^{L(A, \mathbb{R})^{\mathbb{P}_{\max}}} = [U]^V.$$

It is easy to see that $X = \dot{X}^g$ for every $g \in \mathbb{P}_{\max}$ which is generic over $L(A, \mathbb{R})$. We show that $X = [U]$. Suppose $f \in X$, then f itself is a witness for the existential quantifier of the formula

$$\exists h \in X (f \restriction \alpha \subset h)$$

for every $\alpha < \omega_1$. By the homogeneity of $\mathbb{P}_{\max}$,

$$\Vdash_{\mathbb{P}_{\max}}^{L(A, \mathbb{R})} \exists h \in \dot{X} (f \restriction \alpha \subset h)$$

for every $\alpha < \omega_1$. Thus $f \in [U]$. On the other hand, if $f \notin X$, i.e. there is an $s \in C$ such that $s \subset f$, then $f \restriction \text{lh}(s) \notin U$ and hence $f \notin [U]$.

The “moreover” part is given by a standard result in inner model theory which is mainly due to Woodin [26] and Martin–Steel [21];³⁷ the result says that, if there is a proper class of Woodin cardinals, then for every universally Baire set $B \subset \mathbb{R}$,

$$\mathcal{P}(\mathbb{R}) \cap L(B, \mathbb{R}) \subset \Gamma^\infty. \quad \square$$

We prove the tree representation for uB-closed sets.

Theorem 4.20. *Assume Γ^∞ -BMM⁺⁺ and that there is a proper class of Woodin cardinals. Let $1 \leq k < \omega$. Let $X \subset \mathcal{N}_{\omega_1}^k$. The following are equivalent:*

- (1) *X is uB-closed,*
- (2) *for some universally Baire set $A \subset \mathbb{R}$, there exists tree $U \subset (<_{\omega_1} \omega_1)^k$ in $L(A, \mathbb{R})$ such that $X = [U]$.*

Proof. (1) $\implies$ (2) is immediate from Lemma 4.19. We prove (2) $\implies$ (1). Again,

³⁷In the presence of a proper class of Woodin cardinals, the arguments in [26] show that every set of reals in $L(B, \mathbb{R})$ is ∞ -weakly homogeneously Suslin for $B \in \Gamma^\infty$, and the result in [21] says that every ∞ -weakly homogeneously Suslin set of reals is ∞ -homogeneously Suslin.

let us assume that $k = 1$. Let C be the complement of U , i.e.

$$C = \{s \in {}^{<\omega_1}\omega_1 : s \notin U\}.$$

Then $C \in L(A, \mathbb{R})$ and C is universally Baire in the codes as every set of reals in $L(A, \mathbb{R})$ is universally Baire by our assumptions. It is easily verified that

$$X = \mathcal{N}_{\omega_1} \setminus \bigcup \{O_s : s \in C\}.$$

This shows that X is uB-closed. □

Corollary 4.21. *Assume Γ^∞ -BMM $^{++}$ and that there is a proper class of Woodin cardinals. Let $f \in \mathcal{N}_{\omega_1}$. The following are equivalent:*

- (1) $\{f\}$ is uB-closed,
- (2) for some universally Baire set $A \subset \mathbb{R}$, $f \in L(A, \mathbb{R})$,
- (3) for some real $x \in \mathbb{R}$, $f \in L[x]$.

Proof. This is a straightforward application of Theorem 4.20. We first prove (2) $\implies$ (1). Suppose $f \in L(A, \mathbb{R})$ for some universally Baire $A \subset \mathbb{R}$. Let $U = \{s \in {}^{<\omega_1}\omega_1 : s \subset f\}$. Then $U \in L(A, \mathbb{R})$ and $\{f\} = [U]$. By Theorem 4.20, $\{f\}$ is uB-closed. Now we prove (1) $\implies$ (2). Suppose that $\{f\}$ is uB-closed. By Theorem 4.20, let $A \subset \mathbb{R}$ and let $U \in L(A, \mathbb{R})$ be a tree such that $\{f\} = [U]$. Then we have for all $\alpha < \omega_1$ and for all $s \in {}^{<\omega_1}\omega_1$,

$$\begin{aligned} f \restriction \alpha = s &\iff \exists h \in [U] (h \restriction \alpha = s) \\ &\iff \Vdash_{\mathbb{P}_{\max}}^{L(A, \mathbb{R})} \exists h \in [U] (h \restriction \alpha = s), \end{aligned}$$

where the first “ $\iff$ ” is given by $\{f\} = [U]$ and the second “ $\iff$ ” uses the homogeneity of the $\mathbb{P}_{\max}$ forcing. Let

$$\bar{U} = \{s \in {}^{<\omega_1}\omega_1 : \exists \alpha < \omega_1 (\Vdash_{\mathbb{P}_{\max}}^{L(A, \mathbb{R})} \exists h \in [U] (h \restriction \alpha = s))\}.$$

Then $\bar{U} \in L(A, \mathbb{R})$ and it is easily seen that $\bar{U}$ is exactly the set of all initial segments of f . Then $f = \bigcup \bar{U} \in L(A, \mathbb{R})$.

(3) $\implies$ (2) is trivial. As for (2) $\implies$ (3), suppose $f \in L(A, \mathbb{R})$ for some universally Baire $A \subset \mathbb{R}$. Notice that f is a function from ω_1 to ω_1 , thus we may construe f as a subset of ω_1 . Now by Solovay's result, AD implies that every subset of ω_1 is constructible from a real, cf. [17, Lemma 33.14]. This means that there is an $x \in \mathbb{R}$ such that $f \in L[x]$. $\square$

4.2 Σ_1 -Completeness

We introduce two notions of Σ_1 -completeness and show that a set being Σ_1 -complete in either sense implies Π_1 -nondefinability (boldface over H_{ω_2}). We shall work with the generalized Cantor space ${}^{\omega_1}2$ in this section. Recall that a function is continuous if the preimages of open sets are open. It is easy to see that if $f : {}^{\omega_1}2 \rightarrow {}^{\omega_1}2$ is continuous, then there is a unique function $\bar{f} : {}^{<\omega_1}2 \rightarrow {}^{<\omega_1}2$ such that for all $x \in {}^{\omega_1}2$,

$$f(x) = \bigcup_{\xi < \omega_1} \bar{f}(x \restriction \xi).$$

In fact, it is the function defined by

$$\bar{f}(s) = \bigcap_{x \supset s} f(x)$$

for $s \in {}^{<\omega_1}2$. By requiring the function $\bar{f}$ as above to be length-preserving we get the notion of Lipschitz continuity which is a straightforward generalization of Lipschitz functions on $\mathbb{R}$.

Definition 4.22. A function $f : {}^{\omega_1}2 \rightarrow {}^{\omega_1}2$ is *Lipschitz* if there is a length-preserving function $\bar{f} : {}^{<\omega_1}2 \rightarrow {}^{<\omega_1}2$, i.e. $\text{lh}(s) = \text{lh}(\bar{f}(s))$ for all $s \in {}^{<\omega_1}2$, such that for all $x \in {}^{\omega_1}2$,

$$f(x) = \bigcup_{\xi < \omega_1} \bar{f}(x \restriction \xi).$$

The above definition implicitly implies the uniqueness of the function $\bar{f}$. Trivially if a function is Lipschitz then it is also continuous.

Definition 4.23. (i) Let $A \subset {}^{\omega_1}2$ be a set. Then A is Σ_1 -complete if for every $B \subset {}^{\omega_1}2$ which is boldface Σ_1 -definable over H_{ω_2} , there is a continuous

function $f : {}^{\omega_1}2 \rightarrow {}^{\omega_1}2$ such that for all $x \in {}^{\omega_1}2$,

$$x \in B \iff f(x) \in A.$$

- (ii) Let $A \subset {}^{\omega_1}2$ be a set. Then A is *strongly* Σ_1 -complete if for every $B \subset {}^{\omega_1}2$ which is boldface Σ_1 -definable over H_{ω_2} , there is a Lipschitz function $f : {}^{\omega_1}2 \rightarrow {}^{\omega_1}2$ such that for all $x \in {}^{\omega_1}2$,

$$x \in B \iff f(x) \in A.$$

We now describe the construction of the function

$$\text{shift}_1 : {}^{\omega_1}2 \rightarrow {}^{\omega_1}2.$$

Let us first fix a sequence of countable ordinals $(\alpha_i : i < \omega_1)$, which is club in ω_1 , and where the α_i 's are inductively defined by:

- (a) $\alpha_0 = 1$,
- (b) $\alpha_{i+1} = \alpha_i \cdot 2$, where the “ $\cdot$ ” is ordinal multiplication,
- (c) $\alpha_\lambda = \bigcup_{i < \lambda} \alpha_i$, if λ is a limit ordinal.

Now let $x \in {}^{\omega_1}2$ and let I be an interval of ordinals below ω_1 . Write $x \restriction I$ for the sequence $s \in {}^{\text{otp}(I)}2$ such that for every $\xi < \text{otp}(I)$,

$$s(\xi) = x(\min(I) + \xi).$$

Finally we let

$$\text{shift}_1(x) = \bigcup_{i < \omega_1} x \restriction [\alpha_i, \alpha_{i+1})$$

if $\bigcup_{i < \omega_1} x \restriction [\alpha_i, \alpha_{i+1})$ is an element of ${}^{\omega_1}2$; otherwise let $\text{shift}_1(x) = \emptyset$.

Lemma 4.24. *Let $f : {}^{\omega_1}2 \rightarrow {}^{\omega_1}2$ be Lipschitz. Then there is an $x \in {}^{\omega_1}2$ such that*

$$f(x) = \text{shift}_1(x).$$

Proof. Simply let x be any element of ${}^{\omega_1}2$ such that for all $i < \omega_1$,

$$x \restriction [\alpha_i, \alpha_{i+1}) = \bar{f}(x \restriction \alpha_i) \quad {}^{38}$$

where $\bar{f} : {}^{<\omega_1}2 \rightarrow {}^{<\omega_1}2$ is the function as in Definition 4.22. Then

$$\begin{aligned} f(x) &= \bigcup_{i < \omega_1} \bar{f}(x \restriction \alpha_i) \\ &= \bigcup_{i < \omega_1} x \restriction [\alpha_i, \alpha_{i+1}) \\ &= \text{shift}_1(x). \end{aligned}$$

□

Theorem 4.25. *Let $A \subset {}^{\omega_1}2$ be strongly Σ_1 -complete. Then A is not boldface Π_1 -definable over H_{ω_2} .*

Proof. Consider the set

$$\tilde{A} = \{x \in {}^{\omega_1}2 : \text{shift}_1(x) \notin A\}.$$

By the construction of the shift_1 function, $\tilde{A}$ is Σ_1 -definable over H_{ω_2} provided that A is Π_1 -definable over H_{ω_2} . So there is a Lipschitz function f such that

$$x \in \tilde{A} \iff f(x) \in A$$

for all $x \in {}^{\omega_1}2$. By Lemma 4.24, let $x \in {}^{\omega_1}2$ be such that $f(x) = \text{shift}_1(x)$. Then

$$\begin{aligned} x \in \tilde{A} &\iff f(x) \in A \\ &\iff \text{shift}_1(x) \in A \\ &\iff x \notin \tilde{A}. \end{aligned}$$

Contradiction! □

We are able to weaken the assumption of A in the above theorem to Σ_1 -completeness. However, a slight modification of the function shift_1 is needed and its construction will be given in the proof of the follow theorem:

³⁸E.g., we may let x be the concatenation $(0, 0) \frown \bar{f}((0, 0)) \frown \bar{f}((0, 0) \frown \bar{f}((0, 0))) \frown \dots$

Theorem 4.26. *Let $A \subset {}^{\omega_1}2$ be Σ_1 -complete. Then A is not boldface Π_1 -definable over H_{ω_2} .*

Proof. Fix a recursive coding method that codes any sequence $s \in {}^{<\omega_1}2$ into an element $\text{code}(s) \in {}^{\omega_2}2$; we also write $\text{decode}(a)$ for the sequence in ${}^{<\omega_1}2$ which a codes for any $a \in {}^{\omega_2}2$. For $i < \omega_1$, let $\alpha_i = \omega \cdot i$ where “ $\cdot$ ” is ordinal multiplication. Then for each $x \in {}^{\omega_1}2$ and for each $i < \omega_1$, $x \upharpoonright [\alpha_i, \alpha_{i+1})$ is potentially a code for some countable sequence in ${}^{<\omega_1}2$. Let

$$\text{shift}_2(x) = \bigcup_{i < \omega_1} \text{decode}(x \upharpoonright [\alpha_i, \alpha_{i+1}))$$

if $\bigcup_{i < \omega_1} \text{decode}(x \upharpoonright [\alpha_i, \alpha_{i+1}))$ is an element of ${}^{\omega_1}2$; otherwise let $\text{shift}_2(x) = \emptyset$.

Claim 4.27. *Let $f : {}^{\omega_1}2 \rightarrow {}^{\omega_1}2$ be continuous. Then there is an $x \in {}^{\omega_1}2$ such that*

$$f(x) = \text{shift}_2(x).$$

Proof of Claim 4.27. We may recursively define an $x \in {}^{\omega_1}2$ such that

$$\text{decode}(x \upharpoonright [\alpha_i, \alpha_{i+1})) = \bar{f}(x \upharpoonright \alpha_i)$$

for each $i < \omega_1$ as follows. Note that for $i = 0$, $\bar{f}(x \upharpoonright \alpha_0) = \bar{f}(\emptyset)$ only depends on f so we put $\text{code}(\bar{f}(\emptyset))$ as the ω -lengthed initial segment of x . So that $x \upharpoonright \omega$ is defined. Then we concatenate $\text{code}(\bar{f}(x \upharpoonright \omega))$ and $x \upharpoonright \omega$ to get $x \upharpoonright (\omega \cdot 2)$. Eventually we get an $x \in {}^{\omega_1}2$ which satisfies

$$f(x) = \text{shift}_2(x)$$

by repeating this process ω_1 times. □ (Claim 4.27)

Now if A were assumed to be Π_1 -definable over H_{ω_2} , then the set

$$\tilde{A} = \{x \in {}^{\omega_1}2 : \text{shift}_2(x) \notin A\}$$

is Σ_1 -definable over H_{ω_2} . By the Σ_1 -completeness of A , there is a continuous function f such that

$$x \in \tilde{A} \iff f(x) \in A$$

for all $x \in {}^{\omega_1}2$. By Claim 4.27, there is an $x \in {}^{\omega_1}2$ such that $f(x) = \text{shift}_2(x)$. Fix such an x and we get the same contradiction as before:

$$\begin{aligned} x \in \tilde{A} &\iff f(x) \in A \\ &\iff \text{shift}_2(x) \in A \\ &\iff x \notin \tilde{A}. \end{aligned}$$

□

The proof of [10, Theorem 49] actually produces the following theorem.

Theorem 4.28 (Friedman–Hyttinen–Kulikov). *Assume that $V = L$. Then NS_{ω_1} is Σ_1 -complete.*

Proof (sketch). We have the following claim.

Claim 4.29. *Let φ be a Σ_1 -formula and let $X \subset \omega_1$. Then $\varphi(X)$ holds if and only if the set*

$$A_X = \{\alpha < \omega_1 : \exists \beta (L_\beta \models \text{“ZFC}^- \wedge \alpha = \omega_1 \wedge \varphi(X \cap \alpha)\text{”})\}$$

contains a club.

Proof of Claim 4.29. We first show the “ $\implies$ ” direction. It is easy to inductively construct a chain $(M_i : i < \omega_1)$ of countable models such that $(\omega_1^{M_i} : i < \omega_1)$ is a club and for each $i < \omega_1$, there is an elementary embedding $j_i : M_i \rightarrow L_{\omega_2}$. Then for each $i < \omega_1$, by elementarity of j_i , $M_i \models \varphi(X \cap \omega_1^{M_i})$; and by condensation, $M_i = L_{\beta_i}$ for some β_i . This means that $\omega_1^{M_i} \in A_X$ for each $i < \omega_1$.

Now we prove the other direction. Suppose $\neg \varphi(X)$. Fix a club $C \subset \omega_1$ and consider the set

$$D = \{\alpha < \omega_1 : \alpha = \omega_1 \cap \text{Hull}^{L_{\omega_2}}(\alpha \cup \{\omega_1, X, C\})\}.$$

Then $D \subset C$ is also a club. Let α be the ω -th ordinal of D , and by condensation, let $L_{\bar{\beta}}$ be the transitive collapse of $\text{Hull}^{L_{\omega_2}}(\alpha \cup \{\omega_1, X, C\})$. We have

$$L_{\bar{\beta}} \models \text{“ZFC}^- \wedge \alpha = \omega_1 \wedge \neg \varphi(X \cap \alpha)\text{”}.$$

Since φ is a Σ_1 -formula, for any limit ordinal β , if $L_\beta \models \varphi(X \cap \alpha)$, then $\beta > \bar{\beta}$. Now the point is that, $L_{\bar{\beta}+2}$ can already see that $\text{cf } \alpha = \omega$. This means that α is singular in any $L_\beta \models \text{ZFC}^-$ in which $\varphi(X \cap \alpha)$ holds, therefore $\alpha \notin A_X$. $\square$ (Claim 4.29)

Now suppose that A is Σ_1 -definable over L_{ω_2} and let φ be a Σ_1 -formula that defines A . Let f be the function that maps any $X \subset \omega_1$ to $\omega_1 \setminus A_X$. Then by Claim 4.29, f is a continuous function such that

$$X \in A \iff f(X) \in \text{NS}_{\omega_1}$$

for all $X \subset \omega_1$.

$\square$ (Theorem 4.28)

We finish this chapter by proving Lietz's result. We make frequent use of Lemma 2.19 in the proof.

Theorem 4.30 (Lietz [20]). *Assume MM. Then NS_{ω_1} is not Σ_1 -complete.*

Proof (sketch). Let

$$A = \{x \in {}^{\omega_1}2 : x \text{ codes an } \omega_1\text{-tree with a cofinal branch}\}.$$
³⁹

Then A is Σ_1 -definable over H_{ω_2} . We show that there is no continuous function that reduces A to NS_{ω_1} in the sense of Definition 4.23. Suppose otherwise, let $f : {}^{\omega_1}2 \rightarrow {}^{\omega_1}2$ be a continuous function such that $x \in A$ if and only if $f(x) \in \text{NS}_{\omega_1}$ for any $x \in {}^{\omega_1}2$. We sometimes confuse an ω_1 -tree with its code.

Claim 4.31. *Let $\mathbb{P}$ be ω -distributive and stationary set preserving. Then in $V^{\mathbb{P}}$ there is a canonical continuous function $f^+ : ({}^{\omega_1}2)^{V^{\mathbb{P}}} \rightarrow ({}^{\omega_1}2)^{V^{\mathbb{P}}}$ which extends f .*

Proof of Claim 4.31. Fix an $\bar{f} : {}^{<\omega_1}2 \rightarrow {}^{<\omega_1}2$ in V which witnesses the continuity of f , see the discussion in the beginning of this section. For any $x \in ({}^{\omega_1}2)^{V^{\mathbb{P}}}$, let $f^+(x) = \bigcup_{\alpha < \omega_1} \bar{f}(x \restriction \alpha)$. We only need to show that the domain of $f^+(x)$ is ω_1 for any $x \in ({}^{\omega_1}2)^{V^{\mathbb{P}}}$. Suppose otherwise, then there is $p \in \mathbb{P}$ which forces that for some x , $\text{dom}(f^+(x)) \neq \omega_1$. By Lemma 2.19, let $j : M \rightarrow H_\theta$ be an elementary embedding where θ is sufficiently large, M is transitive and of size $\aleph_1$,

³⁹By ω_1 -tree we mean a tree of height ω_1 whose levels are countable.

and $\omega_1 \cup \{\bar{f}, f, p, \mathbb{P}\} \subset \text{ran}(j)$ such that there is an M -generic filter $g \ni j^{-1}(p)$ for $j^{-1}(\mathbb{P})$. But now for any $x \in (\omega_1 2)^{M[g]}$,

$$(f^+)^{M[g]}(x) = \bigcup_{\alpha < \omega_1} j^{-1}(\bar{f})(x \restriction \alpha) = \bigcup_{\alpha < \omega_1} \bar{f}(x \restriction \alpha) = f(x)$$

has domain ω_1 .

□ (Claim 4.31)

Let $\mathbb{P}$ be the ω -closed forcing which adds a Suslin tree and let $\dot{T}$ be the canonical name for the Suslin tree added by $\mathbb{P}$. (Cf. [17, Theorem 15.23].)

Claim 4.32. $f^+(\dot{T})$ is stationary in $V^{\mathbb{P}}$.

Proof of Claim 4.32. Suppose otherwise, let $p \in \mathbb{P}$ force that $f^+(\dot{T})$ is nonstationary in $V^{\mathbb{P}}$. Let $\dot{S} \in V^{\mathbb{P}}$ be the forcing which specializes $\dot{T}$, cf. [17, Theorem 16.17]. By Lemma 2.19, let $j : M \rightarrow H_\theta$ be the same elementary embedding as in Claim 4.31 with $\omega_1 \cup \{\bar{f}, f, p, \mathbb{P}, \dot{S}, \dot{T}\} \subset \text{ran}(j)$ such that there is an M -generic filter $g * h$ for $j^{-1}(\mathbb{P} * \dot{S})$ with $j^{-1}(p) \in g$.

Now “ $f^+(j^{-1}(\dot{T})^g)$ is nonstationary” holds in $M[g]$.⁴⁰ Notice that $(f^+)^{M[g]} = f \restriction M[g]$. So that $f(j^{-1}(\dot{T})^g)$ is nonstationary in V . On the other hand, the Σ_1 statement “ $j^{-1}(\dot{T})^g$ is a special ω_1 -tree” holds in $M[g * h]$. In V , by upward absoluteness, $j^{-1}(\dot{T})^g$ is a special ω_1 -tree and does not have a cofinal branch. Then we have $j^{-1}(\dot{T})^g \notin A$ and hence by Σ_1 -completeness, $f(j^{-1}(\dot{T})^g)$ is stationary in V . This is a contradiction. □ (Claim 4.32)

Claim 4.33. $S \setminus f^+(\dot{T})$ is stationary in $V^{\mathbb{P}}$ for all $S \in \text{NS}_{\omega_1}^{+V}$.

Proof of Claim 4.33. Suppose otherwise, let $p \in \mathbb{P}$ force that $\dot{C} \subset \omega_1$ is club and $\dot{C} \cap S \subset f^+(\dot{T})$. We may think of $\dot{T}$ as a forcing whose order is the reversed tree order, so that $\dot{T}$ is ω -distributive and stationary set preserving in $V^{\mathbb{P}}$, cf. [17, p. 242]. Again, by Lemma 2.19, let $j : M \rightarrow H_\theta$ be the same elementary embedding as in Claim 4.31 with $\omega_1 \cup \{\bar{f}, f, p, \mathbb{P}, \dot{T}, S, \dot{C}\} \subset \text{ran}(j)$ such that there is an M -generic filter $g * h$ for $j^{-1}(\mathbb{P} * \dot{T})$ with $j^{-1}(p) \in g$.

Write $C = j^{-1}(\dot{C})^g$. We have that C is a club with $C \cap S \subset (f^+)^{M[g]}(j^{-1}(\dot{T})^g) = f(j^{-1}(\dot{T})^g)$. Therefore $f(j^{-1}(\dot{T})^g)$ is stationary in V . But “ $j^{-1}(\dot{T})^g$ is an ω_1 -tree with a cofinal branch h ” holds in $M[g * h]$, which means $j^{-1}(\dot{T})^g \in A$.

⁴⁰Here and in what follows, f^+ really is $j^{-1}(f)^+$ from the point of view of $M[g]$. We avoid excessive notations by simply writing f^+ .

By Σ_1 -completeness, $f(j^{-1}(\dot{T})^g)$ is nonstationary in V . This is a contradiction.
 $\square$ (Claim 4.33)

By Claim 4.33 and by the proof of Lemma 4.4, the forcing $\mathbb{P} * \dot{\mathbb{S}} * \dot{\mathbb{R}}$ is stationary set preserving, where $\dot{\mathbb{S}} \in V^{\mathbb{P}}$ is the forcing specializing $\dot{T}$ and $\dot{\mathbb{R}} \in V^{\mathbb{P} * \dot{\mathbb{S}}}$ is the forcing that adds a club through $\omega_1 \setminus f^+(\dot{T})$. Once again by applying Lemma 2.19 to $\mathbb{P} * \dot{\mathbb{S}} * \dot{\mathbb{R}}$, there is an elementary embedding $j : M \rightarrow H_\theta$ which is the same as in Claim 4.31 with $\omega_1 \cup \{\bar{f}, f, \mathbb{P}, \dot{\mathbb{S}}, \dot{\mathbb{R}}, \dot{T}\} \subset \text{ran}(j)$ such that there is an M -generic filter $g * h * k$ for $j^{-1}(\mathbb{P} * \dot{\mathbb{S}} * \dot{\mathbb{R}})$.

We have that “ $f^+(j^{-1}(\dot{T})^g)$ is nonstationary” holds in $M[g * h * k]$, hence also holds in V . On the other hand, $j^{-1}(\dot{T})^g$ is a special ω_1 -tree in $M[g * h]$. Now by the same argument as in Claim 4.32, $f(j^{-1}(\dot{T})^g)$ is stationary in V . Contradiction.
 $\square$ (Theorem 4.30)

Chapter 5

Open questions

It is natural to ask if the assumptions in Theorems 4.1 and 4.16 can be further relaxed. As PFA is already compatible with a Π_1 -definition for NS_{ω_1} , it is unlikely that BMM can be relaxed to BPFA. We are mainly interested in the “ NS_{ω_1} is saturated” part. We ask:

Question 1. Is “ $\text{BMM} + \text{NS}_{\omega_1}$ is precipitous” compatible with a Π_1 -definition of NS_{ω_1} ?

Question 2. Is BMM compatible with a Π_1 -definition of NS_{ω_1} ?

As for Theorem 4.16, we ask if the existence of a Woodin cardinal can be replaced by the saturation of NS_{ω_1} :

Question 3. Assume Γ^∞ -BMM and that NS_{ω_1} is saturated. Is it consistent that NS_{ω_1} is $\text{uB-}\widetilde{\Pi}_1^1$?

We briefly explain the situation of Question 1. One could try to run through the arguments in Chapter 3 and 4 again if a negative answer to Question 1 is expected. However, an important component of Asperó–Schindler forcing will be missing if NS_{ω_1} is only assumed to be precipitous; that is, a generic iteration with $H_{\omega_3}^V$ being the final iterate. Recall that Claverie–Schindler’s result says one can force such an iteration if NS_{ω_1} is precipitous.

Theorem 5.1 (Claverie–Schindler, [5]). *Assume that NS_{ω_1} is precipitous. Then in a stationary set preserving forcing extension of V , there is a generic iteration $(M_i, \pi_{ij} : i < j \leq \omega_1)$ by the nonstationary ideal and its images such that $M_{\omega_1} = H_{\omega_3}^V$.*

We cannot use this theorem to create a situation where Asperó–Schindler forcing is applicable because the forcing given by Theorem 5.1 only works for V ; for our purposes, we need to force over an outer model N of $H_{\omega_3}^V$. The following conjecture would do the work.

Conjecture 1. Assume that NS_{ω_1} is precipitous. Let M be a transitive set such that

- (i) M is elementarily embeddable into H_{ω_3} ,
- (ii) M has size $\aleph_1$, and
- (iii) $\text{cf } \omega_2^M = \text{cf } \omega_3^M = \omega$.

Then there is a stationary set preserving forcing $\mathbb{P}$ which forces a generic iteration $\mathcal{I} = (M_i, \pi_{ij} : i < j \leq \omega_1)$ by the nonstationary ideal and its images such that $M_{\omega_1} = M$.

This conjecture gives us a negative answer to Question 1:

Theorem 5.2. *Assume Conjecture 1. Assume BMM and that NS_{ω_1} is precipitous. Then NS_{ω_1} is not boldface Π_1 -definable over H_{ω_2} .*

Proof. Suppose the theorem is false, let φ_{stat} be a Σ_1 -formula and $X \subset \omega_1$ such that for all $S \subset \omega_1$,

$$S \text{ is stationary} \iff \varphi_{\text{stat}}(S, X).$$

Fix a sufficiently large θ . Let $j : M \rightarrow H_\theta$ be such that

- (a) M is a transitive set of size $\aleph_1$,
- (b) $\text{crit}(j) = \omega_2^M$, and
- (c) $\text{cf } \omega_2^M = \text{cf } \omega_3^M = \omega$.

By Conjecture 1, there is a stationary set preserving forcing $\mathbb{P}$ such that in $V^\mathbb{P}$, there is a generic iteration $\mathcal{I} = (M_i, \pi_{ij} : i < j \leq \omega_1)$ such that $M_{\omega_1} = H_{\omega_3^M}^M$. Then by BMM, such an iteration already exists in H_θ . By exactly the same arguments as in Lemma 4.3, we have that H_θ is a stationary set preserving extension of $H_{\omega_3^M}^M$ such that

- (M.i) $H_\theta \models \exists D \subset \omega_1 (\varphi_{\text{stat}}(D, X) \wedge \forall S \in \text{NS}_{\omega_1}^{+M}, S \setminus D \text{ is stationary}),$
- (M.ii) $H_\theta \models \text{cf } \omega_2^M = \text{cf } \omega_3^M = \omega \wedge H_{\omega_3^M}^M \text{ has size } \aleph_1,$
- (M.iii) $H_\theta \models \text{“NS}_{\omega_1} \text{ is precipitous”},$
- (M.iv) there is a generic iteration $\mathcal{I} = (M_i, \pi_{ij} : i < j \leq \omega_1)$ in H_θ by the nonstationary ideal and its images such that $M_{\omega_1} = H_{\omega_3^M}^M.$

By Shoenfield absoluteness, there is an $\bar{N} \in M^{\text{Col}(\omega, (2^{\aleph_2})^M)}$ such that the above items hold. Now apply the elementarity of j to the above items (M.i) - (M.iv), there is an $N \in V^{\text{Col}(\omega, 2^{\aleph_2})}$ which preserves stationary sets of V such that

- (V.i) $N \models \exists D \subset \omega_1 (\varphi_{\text{stat}}(D, X) \wedge \forall S \in \text{NS}_{\omega_1}^{+V}, S \setminus D \text{ is stationary}),$
- (V.ii) $N \models \text{cf } \omega_2^V = \text{cf } \omega_3^V = \omega \wedge H_{\omega_3}^V \text{ has size } \aleph_1,$
- (V.iii) $N \models \text{“NS}_{\omega_1} \text{ is precipitous”},$
- (V.iv) there is a generic iteration $\mathcal{I} = (M_i, \pi_{ij} : i < j \leq \omega_1)$ in N by the nonstationary ideal and its images such that $M_{\omega_1} = H_{\omega_3}^V.$

By items (V.i)-(V.iii), the formula

$$\Phi \equiv \exists D \subset \omega_1 (\varphi_{\text{stat}}(D, X) \wedge \forall S \in \text{NS}_{\omega_1}^{+V}, S \setminus D \text{ is stationary})$$

is weakly honestly consistent. Now the point is, by item (V.iv), we can find witnesses of the weak honest consistency of Φ that also have a generic iteration with $H_{\omega_3}^V$ being the final iterate, so that the Asperó–Schindler forcing for Φ is stationary set preserving.

We proceed as before. Let $\mathbb{P}_{\text{AS}}$ be a stationary set preserving forcing which forces Φ , i.e.

$$V^{\mathbb{P}_{\text{AS}}} \models \exists D \subset \omega_1 (\varphi_{\text{stat}}(D, X) \wedge \forall S \in \text{NS}_{\omega_1}^{+V}, S \setminus D \text{ is stationary}).$$

Let $\mathbb{S}$ be the forcing that adds a club subset of ω_1 through $\omega_1 \setminus D$. By Lemma 4.4, $\mathbb{P}_{\text{AS}} * \mathbb{S}$ is stationary set preserving. We have

$$V^{\mathbb{P}_{\text{AS}} * \mathbb{S}} \models \exists D (\varphi_{\text{stat}}(D, X) \wedge D \text{ is nonstationary}).$$

By BMM,

$$V \models \exists D (\varphi_{\text{stat}}(D, X) \wedge D \text{ is nonstationary}),$$

a contradiction!

□

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